

[B17 BS 1101]

I B. Tech I Semester (R 17) Regular Examinations

ENGLISH-I

**(Common to all branches)
MODEL QUESTION PAPER**

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. A) Correct the following sentences. (7M)
- The machineries were expensive.
 - Suppose, if you arrive late, you will miss the show.
 - Choose the best of the two options.
 - I enjoyed during the holidays.
 - I have seen him yesterday.
 - The teacher gave us many advices.
 - My dog is better than him.
- B) a) Write appropriate quantifiers for each sentence. (3M)
- The project is complicated than the last one
 - I have to buypairs of blue and black jeans soon.
 - There is no water in the bottle.
- b) Re-write the sentences by using Gerunds, to-infinitives or plain infinitive forms. (4M)
- I noticed him (write) picture postcards.
 - I feel happy to be..... (sing) a song.
 - They felt surprised to (select) by the manager.
 -(Garden) is a pleasant activity.
- (OR)
2. A) Fill in the blanks using the appropriate forms of verbs given in the brackets. (7M)
- In a fit of rage, she ____ up the letter. (Tear)
 - We couldn't have ____ a better day for organizing the party. (Choose)
 - It's high time you ____ your mistake. (Realise)
 - The poem 'The Gift of India' ____ (write) by Sarojini Naidu in 1915.
 - We ____ for five years now. (marry)
 - When I ____ home, I found that there was no edible oil left. (go)
 - The Journalist reported that the miscreants ____ a havoc in the city. (create)
- B) a) Fill in the blanks by using appropriate conjunctions (3M)
- Receptionists must be able to relay information ____ pass messages accurately.
 - Mary is a member of the Historical Society ____ the Literary Society.
 - My friend didn't work hard ____ he got through the exam.
- b) Punctuate the following sentences. (4M)
- sunil sharma is documentation development manager at cerner corporation one of the world's largest medical software developers
 - As part of his job Sunil writes web-based content for Cerner.
 - Hang him not leave him.
 - my friend suresh who is in bengaluru has come today.

UNIT-II

3. A) Write one word substitutions to the following and write sentences by using them. Marks will be awarded only when both the points are correctly answered. (7M)
- Language which is confusing and unintelligible.
 - One who prepares plans for buildings.
 - A great lover of books
 - A person in charge of a museum
 - A man who thinks only for himself
 - One who kills animals and sells their flesh
 - A person with a long experience in a specific field
- B) a) Give synonyms for the following words and use them in your own sentences. (3M)
- Euphoria
 - Vicious
 - Ostentatious
- b) Give antonyms for the following words and use them in your own sentences. (4M)
- Truce
 - Terse
 - Supercilious
 - Emerge

(OR)

4. A) Give meanings for the following idioms and also use them in your own sentences. (7M)
- The cream of the crop
 - An arm and a leg
 - Hand in glove
 - Hue and cry
 - Hard and fast
 - Explore all avenues
 - Spill the beans
- B) a) Give synonyms for the following words and use them in your own sentences. (3M)
- Sacrilege
 - Pugnacious
 - Vitiate
- b) Give antonyms for the following words and use them in your own sentences. (4M)
- Succinct
 - stigmatize
 - recalcitrant
 - Adamant

UNIT-III

5. A) Read the following paragraph and answer the questions: (7M)

The study of history provides many benefits. First, we learn from the past. We may repeat mistakes, but, at least, we have the opportunity to avoid them. Second, history teaches us what questions to ask about the present. Contrary to some people's view, the study of history is not the memorization of names, dates, and places. It is the thoughtful examination of the forces that have shaped the courses of human life. We can examine events from the past and then draw inferences about current events. History teaches us about likely outcomes.

Another benefit of the study of history is the broad range of human experience which is covered. War and peace are certainly covered as are national and international affairs. However, matters of culture (art, literature, and music) are also included in historical study. Human nature is an important part of history: emotions like passion, greed, and insecurity have influenced the shaping of world affairs. Anyone who thinks that the study of history is boring has not really studied history.

- What is the central idea of this passage?
 - In the first paragraph, 'inferences' mean?
 - Which method of teaching history would the author of this passage support?
 - In the second paragraph, 'shaping of world affairs' Means.
 - What is the conclusive thought of the author?
 - Give an appropriate title for the written discourse.
 - How reliable is the written history; and/or is it just 'his' story?
- B) Develop a paragraph (200 words) based on the following hints and provide an appropriate title for the same. (7M)

As the 11th President of India---- the Indian National Congress-----
'people's president', he was----- His contribution -----Bharat Ratna. During
-----in India. He is the -----India: 2020 and Ignited Minds.

(OR)

6. A) Read the following paragraph and answer the questions: (7M)

Work expands so as to fill the time available for its completion. The general recognition of this fact is shown in the proverbial phrase, 'It is the busiest man who has time to spare.' Thus, an elderly lady at leisure can spend the entire day writing a postcard to her niece. An hour will be spent in finding the postcard, hunting for spectacles, half an hour to search for the address, an hour and a quarter in composition and twenty minutes in deciding whether or not take an umbrella when going to the pillar box in the street. The total effort that would occupy a busy man for three minutes, all told, may in this fashion leave another person completely exhausted after a day of doubt, anxiety and toil.

1. What happens when the time to be spent on some work increases?
2. Explain the sentence: 'Work expands so as to fill the time available for its completion.'
3. Who is the person likely to take more time to do work?
4. What is the total time spent by the elderly lady in writing a postcard?
5. What does the expression 'pillar box' stand for?
6. Suggest an appropriate title for the passage.
7. 'It is the busiest man who has time to spare' Elaborate the semantic content of it.

- B) Develop a paragraph (200 words) based on the following hints and provide an appropriate title for the same. (7M)

_____not luck but labor _____ Luck _____ever waiting_____ ;labour _____ strong-will turns up something. Luck _____ news of a legacy; labour _____ the foundation of competence. Luck _____on chance, labour _____ character.

UNIT-IV

7. A) Write an Essay on 'Terrorism, a social evil' (7M)
B) Draft an E-Mail to your friend about your career plans. (7M)

(OR)

8. A) Write an essay on 'Facing a book vis-à-vis Facebook' (7M)
B) Present an argument in about 250 words on 'Technology replacing Teachers'. Substantiate your argument with reasons. (7M)

UNIT-V

9. A) Write a feasibility report on 'Setting up a Water / Power Unit at your campus.' (7M)
ii. Write a report on Educational Tour

- B) Draft a pamphlet on any Electronic home appliances/Places of tourists' interest/an Educational institution/ an exhibition. (7M)

(OR)

- 10.A) Write a feasibility report on 'Educational Tour'. (7M)
B) Write a letter to a renowned person, requesting him to be the Chief Guest for the cultural festival of your college. (7M)

[B17 BS 1101]

[B17 BS 1102]
I B. Tech I Semester(R17) Regular Examinations
MATHEMATICS-I
(Common to all branches)

MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.
All questions carry equal marks.

UNIT – I

1. (a) Solve $\frac{dy}{dx} + (\tan x)y = (\sec x)y^3$. (7M+7M)
(b) Find the orthogonal trajectories of the family of parabolas $ay^2 = x^3$.

(OR)

2. (a) Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$. (7M+7M)
(b) A body originally at 80°C cools down to 60°C in 20 minutes, the temperature of air being 40°C . What will be the temperature of the body after 40 minutes from the original?

UNIT - II

3. (a) Solve $(D^3 - D)y = 2x + 1 + 4 \cos x$. (7M+7M)
(b) solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \log x$ by the method of variation of parameters.

(OR)

4. (a) solve $(D^2 + 3D + 2)y = e^{e^x}$. (7M+7M)
(b) The differential equation for a circuit in which self inductance and capacitance neutralize each other is $L \frac{d^2i}{dt^2} + \frac{i}{C} = 0$. Find the current i as a function of t , given that i is maximum current and $i = 0$ when $t = 0$.

UNIT - III

5. (a) Find $L\{t \cos at\}$ and $L\left\{\int_0^t e^{-t} \cos t \, dt\right\}$. (7M+7M)
(b) Using convolution theorem evaluate $L^{-1}\left\{\frac{1}{(s+a)(s+b)}\right\}$.

(OR)

6. (a) Find $L^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\}$. (7M+7M)

- (b) Solve $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = e^{-t}$, $y(0) = y'(0) = 1$ by using Laplace transforms.

UNIT – IV

7. (a) If $U = \tan^{-1} \frac{x^3 + y^3}{x - y}$ and $x U_x + y U_y = \sin 2U$, prove that

$$x^2 U_{xx} + 2xy U_{xy} + y^2 U_{yy} = 2 \cos 3U \sin U. \quad (7M+7M)$$

- (b) If $u = x^2 - 2y^2$, $v = 2x^2 - y^2$ where $x = r \cos \theta$, $y = r \sin \theta$

show that $\frac{\partial(u,v)}{\partial(r,\theta)} = 6r^3 \sin 2\theta$.

(OR)

8. (a) Expand $x^2y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$ using Taylor's theorem.

(7M+7M)

- (b) By using the method of differentiation under the integral sign

prove that $\int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)} dx = \frac{\pi}{2} \log(1+a)$, $a \geq 0$.

UNIT – V

9. (a) Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$.

(7M+7M)

(b) solve $(D^2 - DD' - 2D'^2)z = (y - 1)e^x$.

(OR)

10. (a) Solve $x(y - z)p + y(z - x)q = z(x - y)$.

(7M+7M)

(b) solve $(D + D' - 1)(D + 2D' - 3)z = 3x + 6y + 4$.

[B17 BS 1103]
I B. Tech I Semester (R 17) Regular Examinations
MATHEMATICS-II
(Mathematical Methods)
 (Common to CSE,ECE & IT)

MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT - I

- 1.a) Find a root of $x^3 - x - 11 = 0$ using the bisection method correct to three decimal places. (7M+7M)
 b) Find the cube root of 41 using Newton-Raphson method.
(OR)
 2. a) Find a real root of the equation $x \log_{10} x = 1.2$ by Regula-false method correct to three decimal places. (7M+7M)
 b) Find the positive root of the equation $3x = \cos x + 1$ by iteration method.

UNIT - II

3. a) Using Gauss forward difference formula, Find Y (8), from the following table (7M+7M)

X	0	5	10	15	20	25
Y	7	11	14	18	24	32

- b) Find the interpolating polynomial $f(x)$ for the data of the following table

x	0	1	4	5
f(x)	4	3	24	39

(OR)

4. a) Using Gauss backward formula, find $f(42)$, from the following table (7M+7M)

X	20	25	30	35	40	45
f(x)	354	332	291	260	231	204

- b) Using Lagrange's interpolation formula find Y(10) from the following table

x	5	6	9	11
Y	12	13	14	16

UNIT - III

5. a) Evaluate $\int_0^2 \frac{dx}{x^3 + x + 1}$ by using Simpsons $1/3^{\text{rd}}$ rule with $h = 0.25$ (7M+7M)
 b) Evaluate $y(0.8)$ using Runge Kutta method given $y' = (x + y)^{\frac{1}{2}}$, $y(0.4) = 0.41$

(OR)

6. a) A rocket is launched from the ground. Its acceleration $a(t)$ measured every 5 seconds is tabulated below. Use trapezoidal rule to find the velocity and the position of the rocket at $t = 40$ seconds. (7M+7M)

t	0	5	10	15	20	25	30	35	40
a(t)	40.0	45.25	48.50	51.25	54.35	59.48	61.5	64.3	68.7

- b) Given $y' = x + \sin y$, $y(0) = 1$, compute $y(0.2)$ and $y(0.4)$ with $h = 0.2$ using modified Euler's method.

UNIT – IV

7. a) Find the Fourier series to represent $f(x) = x - x^2$ from $x = -\pi$ to $x = \pi$. (7M+7M)
 b) Obtain the sine series for $f(x) = x$ in $0 \leq x \leq \pi$.

(OR)

8. a) Obtain the Fourier series for the function $f(x) = \begin{cases} \pi x, & 0 \leq x < 1 \\ \pi(2 - x), & 1 \leq x \leq 2 \end{cases}$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$. (7M+7M)
 b) Find half range cosine series for $f(x) = x(2 - x)$ in $0 < x < 2$.

UNIT – V

9. a) Find the Fourier Transform of $\frac{1}{\sqrt{|x|}}$. (7M+7M)

- b) Find the Fourier integral representation for $f(x) = \begin{cases} 1 - x^2, & \text{for } |x| \leq 1 \\ 0, & \text{for } |x| > 1 \end{cases}$

(OR)

10. a) Find the inverse Fourier transform $f(x)$ of $F_s(p) = \frac{p}{1+p^2}$. (7M+7M)

- b) Find the Fourier cosine transform of e^{-ax} . Hence evaluate $\int_0^{\infty} \frac{\cos \lambda x}{x^2 + a^2} dx$

[B17 BS 1104]
I B. Tech I Semester (R 17) Regular Examinations
ENGINEERING PHYSICS
(Common to CSE,ECE & IT)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT – I

1. (a) Describe the interference phenomena In thin transparent films for reflected light and obtain the conditions for maxima and minima. [7M]
(b) Discuss the Fraunhofer diffraction of monochromatic light at a single slit. [7M]

(OR)

2. (a) Describe, with a neat sketch, the design and working of Michelson's interferometer. [7M]
(b) Explain how the resolving power of a grating can be determined. [7M]

UNIT – II

3. (a) Differentiate spontaneous and stimulated emission processes and obtain the Einstein's relation for spontaneous to stimulated emission coefficients. [7M]
(b) Define numerical aperture of an optical fiber and derive an expression for the same. [7M]

(OR)

4. (a) With neat sketches, explain the principle and working of He – Ne gas laser system. [7M]
(b) Explain the characteristics of lasers and mention the applications of lasers. [7M]

UNIT – III

5. (a) Discuss the electric fields induced due to time varying magnetic fields and deduce the Faraday's law. [7M]
(b) Describe any one method of detecting ultrasonics and mention the important applications of Ultrasonics. [7M]

(OR)

6. (a) Explain the concept of displacement current, and describe the significance of Maxwell's equations. [7M]
(b) What is magnetostriction and describe the magnetostriction method of producing Ultrasonics. [7M]

UNIT – IV

7. (a) What are matter waves and describe an experiment confirming the wave nature of electrons. [7M]
(b) What are the salient features of Kronig - Penny model. [7M]

(OR)

8. (a) Explaining the physical significance of wave function of a particle derive the Schrodinger's time independent wave equation. [7M]
(b) Using band theory of solids how do you classify the materials. [7M]

UNIT - V

9. (a) What is a unit cell and describe the different crystal systems possible in solids. [7M]
(b) What are nano materials and explain the chemical vapour deposition method of fabricating nano materials. [7M]
- (OR)
10. (a) Define packing fraction and deduce the packing fraction for a simple cubic structure. [7M]
(b) Define the basic approaches of fabricating nano materials and discuss the sol-gel method. [7M]

[B17 BS 1105]
I B. Tech I Semester (R 17) Regular Examinations
ENGINEERING CHEMISTRY
(Common to CIV, EEE & ME)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT - I

1. (a) Explain the mechanism of free radical Polymerization reaction with a suitable example. [7M]
(b) Distinguish between thermoplastic and thermosetting resins. [7M]

(OR)

2. (a) What are conducting Polymers? Discuss the applications of conducting Polymers. [7M]
(b) Write notes on Bu Na – S and Bu Na – N. [7M]

UNIT - II

3. (a) Explain the Proximate analysis of coal and give its significance. [7M]
(b) Explain the fractional distillation of crude oil. [7M]

(OR)

4. (a) Write notes on (i) Knocking (ii) CNG [7M]
(b) How Synthetic Petrol can be prepared by Bergius Process. [7M]

UNIT - III

5. (a) Explain the mechanism of electrochemical theory of corrosion with neat diagram. [7M]
(b) Describe briefly about cathodic Protection. [7M]

(OR)

6. (a) Explain Hydrogen – Oxygen fuel cell with neat cell diagram [7M]
(b) Discuss on various constituents of Paint. [7M]

UNIT - IV

7. (a) What is hardness? How it is determined by EDTA method? Explain. [7M]
(b) Describe with equations how water can be softened using Lime & Soda Process [7M]

(OR)

8. (a) Discuss various sterilizing methods used in municipal water treatment. [7M]
(b) Illustrate the reverse osmosis Process with a neat diagram. [7M]

UNIT - V

9. (a) Discuss chemistry involved in setting and hardening of cement? [7M]
(b) What are refractories? Discuss the classification of refractories. [7M]

(OR)

10. (a) Write the engineering applications of Liquid Crystals. [7M]
(b) Explain the stoichiometric defects in crystals. [7M]

[B17 BS 1105]

[B17 CS 1101]
I B. Tech I Semester (R 17) Regular Examinations
COMPUTER PROGRAMMING USING C
(Common to CSE,ECE & IT)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. a) Discuss about computer languages. [7M]
b) Explain c tokens. [7M]

(OR)

2. a) Explain different c operators. [7M]
b) write about algorithm, pseudo code and flowchart. [7M]

UNIT-II

3. a) Discuss various looping techniques in c. [7M]
b) Write a c program for summation of n numbers. [7M]

(OR)

4. a) Explain 2-D arrays and character arrays in c. [7M]
b) Write a c program to find frequency of characters of a string. [7M]

UNIT-III

5. a) Explain parameter passing techniques in c. [7M]
b) Write a c program for towers of Hanoi using recursive function. [7M]

(OR)

6. a) Explain storage classes in c. [7M]
b) Write a c program for Fibonacci series using recursive function. [7M]

UNIT-IV

7. a) What is a pointer? How pointer variables are initialized. [7M]
b) Write a program to print command line arguments on the screen. [7M]

(OR)

8. a) Discuss character pointers with examples. [7M]
b) Write a c program to pass pointer variables as function arguments. [7M]

UNIT-V

9. a) Explain the difference between structure and union and write a program to find sum of marks in 3 subjects for a student using structures. [7M]
b) Explain different bit-wise operators in c. [7M]

(OR)

10. a) Explain about the input and output operations of a file. [7M]
b) Write a c program to open a file and to print its contents on screen. [7M]

[B17 CS 1101]

[B17 CE 1101]
I B. Tech I Semester (R 17) Regular Examinations
ENVIRONMENTAL STUDIES
(Common to all Branches)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.
All questions carry equal marks.

UNIT-I

- 1 a). Define Environment. Write the scope and importance of the environmental studies. [7M]
b). Elucidate the concept of Global Environmental crisis. [7M]

(OR)

- 2 a). What is an ecosystem? Write the structure and functions of an ecosystem. [7M]
b). Write a brief note on forest resources. [7M]

UNIT-II

- 3 a). Describe the values of Biodiversity. [7M]
b). Write about in-situ and ex-situ conservation. [7M]

(OR)

- 4 a). Describe Biogeographical Classification of India. [7M]
b). India as a mega-diversity habitat – Explain [7M]

UNIT-III

- 5 a). Effects of modern agriculture on land. [7M]
b). What are the benefits and problems of dams? [7M]

(OR)

- 6 a). Write about floods and droughts? [7M]
b). Discuss the impact of energy use on environment. [7M]

UNIT-IV

- 7 a). What are the causes, effects and control measures of air pollution? [7M]
b). What is solid waste management? Explain its methods. [7M]

(OR)

- 8 a). Elucidate the results of population growth on environment? [7M]
b). Write notes on Rain water harvesting with a neat sketch [7M]

UNIT-V

- 9 a). Mention the different environmental acts and write about one. [7M]
b). Write notes on Environmental impact Assessment. [7M]

(OR)

- 10 a). Write short notes on any two environmental case studies. [7M]
b). Write a report on a visit to an environmental polluted area? [7M]

[B17 CE 1101]

[B17 ME 1101]
I B. Tech I Semester (R 17) Regular Examinations
ENGINEERING MECHANICS
(Common to CIV,EEE & ME)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

Assume the missing data if any, suitably

UNIT-1

1. (a) State and prove Varignon's theorem. [7 M]
(b) Two cylinders of diameter 100 mm and 50 mm, weighing 200 N and 50 N, respectively are placed in a trough as shown in Figure 1. Assuming smooth surfaces, find the reactions at the points of supports 1, 2, 3 and 4. [7 M]

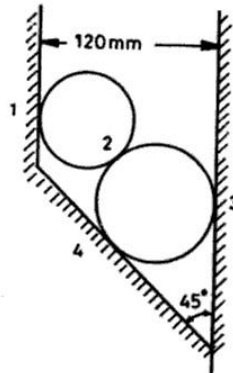


Figure 1

[OR]

- 2 (a) A string ABC of length l carries a small pulley C from which a Load W is suspended as shown in Figure 2. The string hangs between two vertical walls which are at a distance d apart. The end A is higher than the end B by height h . Find the position of equilibrium defined by the angle α . Assume $d = l/2$ and $h = l/4$. [7 M]

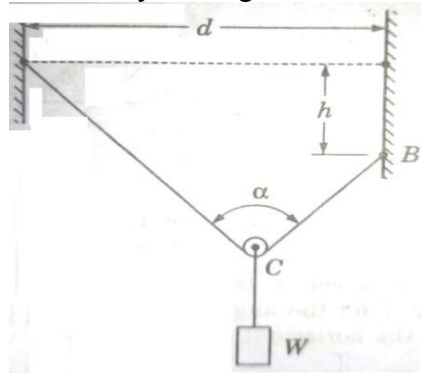


Figure 2

- (b) Two identical prismatic bars AB & CD each weighing 200 N are welded together to form a Tee and are suspended in a vertical plane as shown in Figure 3. Calculate the values of the θ that the bar AB will make with the vertical when a vertical load of 200 N is applied at D. [7 M]

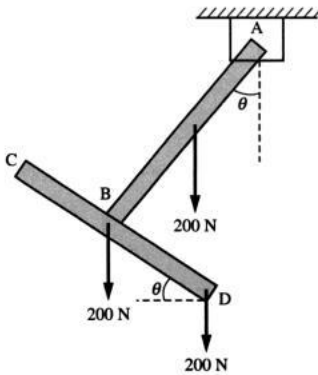


Figure 3

UNIT-II

- 3 (a) Derive the centroid of a wire bend in the form of a sector of an arc by taking the radius as 'r' and angle of sector as ' θ '. [7 M]
 (b) Determine the centroid of the shaded segment for Figure 4 by taking $a = 18$ m and $\alpha = 45^\circ$. [7 M]

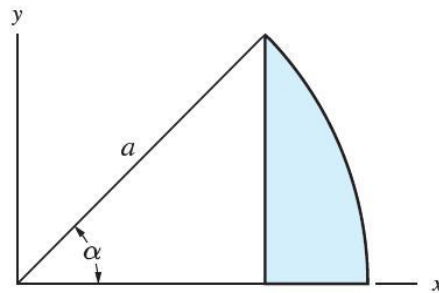


Figure 4

[OR]

- 4 (a) Derive the moment of inertia of triangle about its centroidal axis and also deduce the same about its base. [7 M]
 (b) Determine the moment of Inertia of the T-section shown in Figure 5 about its centroidal axis. [7 M]

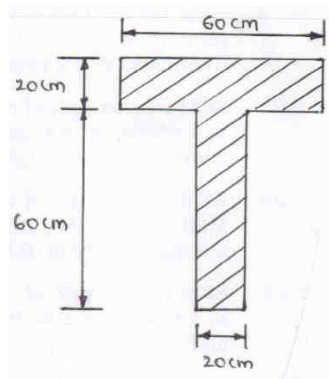


Figure 5

UNIT-III

- 5 (a) Explain the terms angle of repose, cone of friction and write the laws of friction. [7 M]
 (b) Referring to the Figure 6 given above, determine the least values of the force P to cause motion to impend right wards. Assume the coefficient of friction under the blocks to be 0.2 and pulley to be frictionless. [7 M]

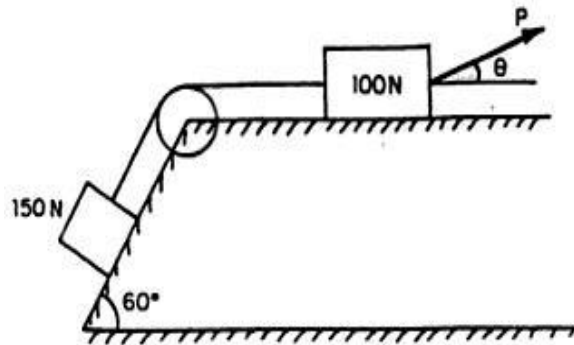


Figure 6

[OR]

- 6 (a) A uniform ladder 5m long on a horizontal ground and leans against a smooth vertical wall at an angle of 70° with the horizontal. The weight of the ladder is 90 N and acts at its middle. The ladder is at the point of sliding, when a man weighing 75N stands on a rung 3.5m from the top of the ladder. Calculate the co-efficient of the friction between the ladder and the floor. [7 M]

- (b) Find out the forces in all the members of a pin jointed truss as shown in Figure 7 by using method of Joints. [7 M]

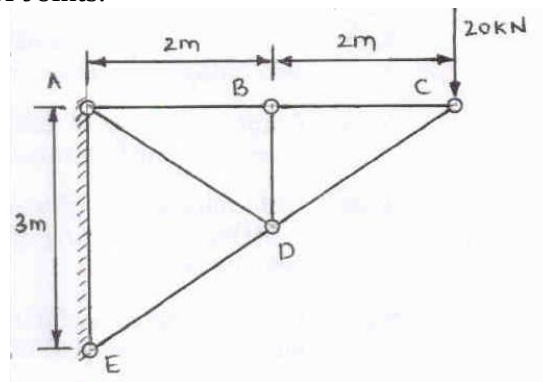


Figure 7

UNIT-IV

- 7 (a) A stone is dropped from the top of a tower 60 m high. At the same instant, another stone is thrown vertically upwards from the foot of tower to meet the first stone at a height of 18 m. Determine (i) the time when the two stones meet; (ii) the velocity with which the second stone was thrown up. [7 M]
 (b) Weight W and $2W$ are supported in a vertical plane by a string and pulleys arranged as shown in Figure 8. Find the magnitude of an additional weight Q applied on the left which will give a downward acceleration $a = 0.1g$ to the weight W . [7 M]

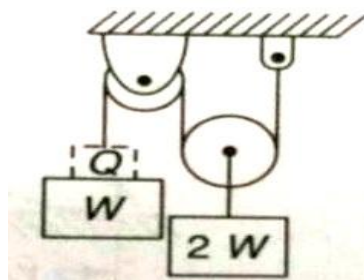


Figure 8

[OR]

- 8 (a) Define Time of Flight, Range and Maximum Height of a projectile. [7 M]
 (b) Derive the general equation of projectile motion. [7 M]

UNIT-V

- 9 (a) A flywheel is rotating at 150 R.P.M. and after 8 seconds it is rotating at 120 R.P.M.. If the retardation is uniform, determine number of revolutions made by the flywheel and the time taken by the flywheel before it comes to rest from the speed of 150 R.P.M. [7 M]
 (b) A rotor of weight $W = 1720$ N and radius of gyration $k = 100$ mm is mounted on a horizontal shaft and set in rotation by a falling weight $W = 1720$ N as shown in Figure 9. If the system is released from rest, find the velocity of the block after it has fallen through a distance of 3 m. [7 M]

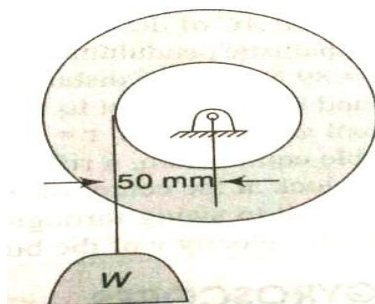


Figure 9

[OR]

- 10 (a) A body is rotating with an angular velocity of 8 radian/s. After 5 seconds, the angular velocity of the body becomes 28 radian/s. determine the angular acceleration of the body. [7 M]
 (b) Three bodies, a sphere, a cylinder and a hoop each having the same mass and radius are released from rest from an inclined plane of angle θ . Determine the velocity of each of the bodies after it has rolled down the incline plane through a distance s . [7 M]

[B17 ME 1102]
I B. Tech I Semester (R 17) Regular Examinations
ENGINEERING DRAWING
(Common to CIV,EEE & ME)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

Assume the missing data if any, suitably

UNIT-I

1. An inelastic string 145 mm long has its one end attached to the circumference of a circular disc of 40 mm diameter. Draw the curve traced out by the other end of the string, when it is completely wound around the disc, keeping the string always tight. [14 M]

[OR]

2. Two fixed points A and B are 100mm apart, Trace the complete path of a point P moving (in the same plane as that of A and B) in such a way that the sum of its distance from A and B is always the same and equal to 125mm. Name the curve and draw another curve parallel to and 25mm away from this curve. [14 M]

UNIT-II

3. A line AB, of 80 mm long has its end A, 15 mm in front of VP and 20 mm above HP. The other end B is 40 mm above HP and 50 mm in front of VP. Draw the projections of the line and determine the inclinations of the line with HP and VP. [14 M]

[OR]

4. (a) Draw the projections of a 75mm long straight line in the following positions: (i) parallel to and 30mm above the HP and in the VP; (ii) perpendicular to the VP, 25mm above the HP and its one end in the VP; (iii) Inclined at 30° to the HP and its one end 20mm above it, parallel to and 30mm in front of the VP. [7 M]

(b) Draw the projections of the following points on the same ground line, keeping the projectors 25mm apart. (i) Point A in the HP and lying 20mm behind the VP; (ii) Point B is 40mm above the HP and 25mm in front of the VP; (iii) Point C is 25mm below the HP and 25mm behind the VP; (iv) Point D is 15mm above the HP and 50mm behind the VP. [7 M]

UNIT-III

5. Draw a rhombus of diagonals 100 mm and 60 mm long, with the longer diagonal horizontal. The figure is the top view of a square of 100mm long diagonals, with a corner on the ground. Draw its front view and determine the angle which its surface makes with the ground. [14 M]

[OR]

6. A semicircular plate of 40mm diameter has its straight edge in the VP and inclined at 45° to the HP, the surface of the plate makes an angle of 30° with the VP. Draw its projections. [14 M]

UNIT-IV

7. A hexagonal pyramid, base 25mm side and axis 50mm long, has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the VP. Draw its projections. [14 M]

[OR]

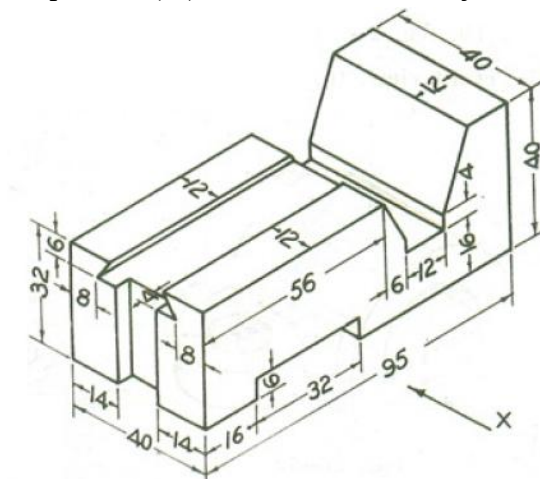
8. Draw the projections of a cylinder 75mm diameter and 100mm long, lying on the ground with its axis inclined at 30° to the VP and parallel to the ground. [14 M]

UNIT-V

9. A square pyramid with base side 40mm and height 60mm is resting on a cube of sides 50mm, the axes of the cube and the pyramid being in the same line. Two sides of the base of the pyramid are parallel to the edges of the cube. Draw the isometric view. [14 M]

[OR]

10. Draw (i) Front View (ii) Top View (iii) Side View of the object shown below: [14 M]



All the dimensions are in mm

[B17 BS 1201]
I B. Tech II Semester (R 17) Regular Examinations
ENGLISH-II
(Common to all Branches)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. A. i) Describe how education is the greatest resource. (4M)
ii) Write a brief note on the great contribution made by Kalam to the science and technology. (3M)
B. Imagine that you are a builder and draft a letter of tender quotation to the Chief Engineer of Department of Roads and Buildings of Karnataka for constructing an administrative building. (7M)
- (OR)**
2. A. i) What, according to the author, is the source of problems for civilizations? (4M)
ii) Who had most influenced the value system of Kalam when he was young? (3M)
B. Imagine that you are the managing Director of a big company that manufactures electronic goods like Music systems, DVDs, LCDs. Write a business letter addressed to the Board of directors requesting them to attend a meeting to be held in the ensuing month. Give the agenda of the meeting also. (7M)

UNIT-II

3. A. i) What is the layman's view of atomic bomb? How right is he in thinking so? Who do you think is to be held responsible for the destruction created by technology? Support your opinion with suitable examples. (4M)
ii) What were some of the changes that Raman had initiated at the Indian Institute of Science? (3M)
B. Make notes on the following passage. (7M)
Here is an excerpt from one of Abdul Kalam's essays.
Knowledge has many forms and it is available at many places. It is acquired through education, information, intelligence and experience. It is available in academic institutions with teachers, in libraries, in research papers, seminar proceedings and in various organizations and work places with workers, managers, in drawings, in process sheets and on the shop floors. Knowledge, though closely linked to education, comes equally from learning skills, such as those possessed by our artists, craftsmen, hakims, vaidyas, philosophers and saints, as also our housewives. Knowledge plays a very important role in their performance and output too. Our heritage and history, the rituals, epics and traditions that form part of our consciousness are also vast resources of knowledge as are our libraries and universities. There is an abundance of unorthodox, earthy wisdom in our villages. There are hidden treasures of knowledge in our environment, in the oceans, bio-reserves and deserts, in the plant and animal life. Every state in a country has a unique core competence for a knowledge society

(OR)

4. A. i) Describe any modern invention with its positive and negative effects on the society. (4M)
ii) List out the awards and achievements of Sir C.V. Raman. (3M)
B. Make notes on the following passage. (7M)
It is not luck but labor that makes a man. Luck, says an American writer, is ever waiting for something to turn up; labour with keen eyes and strong will always turns up something. Luck lies in bed and wishes the postman would bring him news of a legacy; labour turns out at six and with busy pen and ringing hammer lays the foundation of competence. Luck whines, labour watches. Luck relies on

chance, labour on character. Luck slips downwards to self-indulgence; labour strides upwards and aspires to independence. The conviction, therefore, is extending that diligence is the mother of good luck. In other words, that a man's success in life will be proportionate to his efforts, to his industry, to his attention to small things.

UNIT-III

5. A.(i)How should one avoid culture shock before experiencing it when one goes to a new place? What precautions would help in living peacefully in a new place of new culture? (4M)
 (ii)Explain in brief Baba's theory on the hitting of cosmic rays on earth's atmosphere (3M)

- B. Write a paragraph on one of the following ideas. (7M)
 i) Facebook ii) Barking dog seldom bites.

(OR)

- 6.A i) How does a person become a cultural entity ? (4M)
 ii) Imagine that you have been elected as the Cultural Secretary of the Students' Association and you have to give a ten-minute speech outlining your plans for the academic year. Write out your speech in about 75 words. (3M)
 B. Write an essay on Homi Bhabha's life and his academic and professional journey. (7M)

UNIT-IV

7. A i) How does Shirley Jackson trivialize the grave practice of the communities traditional stoning and what message might Jackson be trying to convey to the reader through the treatment of the characters' behavior? (4M)
 ii) What were two types of services devised by the British in the Indian Education Services? Why? (3M)

- B. Rewrite the following sentences correcting the errors: (7M)
 i. He plays football when he was free
 ii. He drunk coffee everyday when he was young
 iii. Had your breakfast in the morning?
 iv. He drunk coffee everyday when he was young.
 v. Had your breakfast in the morning?
 vi. Why haven't you been along with me for the event last month?
 vii. Never I have seen such a person!

(OR)

8. A. i) What is black box? Who made it? When and why is it significant? (4M)
 ii) Fill in the blanks with appropriate prepositions. (3M)
 a. She was senior___ me when we were with the academic project _____some time.
 b. One who believes ___ and a devotee ___ God is a theist.
 c. He is angry___ her behavior as she always lies ___ him.
 d. Write an essay on the contribution of J.C. Bose to the field of science. (7M)

UNIT-V

9. A. i) How did the relationship between Microsoft and IBM begin? (4M)
 ii) Collocate the given words of the list A with those of the list B.

A	B
i. Resounding	enemies ()
ii. Bitter	success ()
iii. Death	blow ()

- B. Write an essay on Dr. Prapulla Chandra Ray's life and his academic and professional journey (7M)

(OR)

10. A i) Describe How Gates worked for the development of Microsoft. (4M)
ii) Describe the compound Ray discovered. What are its properties? (3M)
B. Write a business report on 'Setting up a Pharmaceutical Lab and Manufacturing Unit at Visakhapatnam, Andhra Pradesh. (7M)

[B17 BS 1202]
I B. Tech II Semester (R 17) Regular Examinations
MATHEMATICS-II
(Mathematical Methods)
(Common to CIV, EEE & ME)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT - I

- 1.a) Find a root of $x^3 - x - 11 = 0$ using the bisection method correct to three decimal places. (7M+7M)
 b) Find the cube root of 41 using Newton-Raphson method.
(OR)
 2. a) Find a real root of the equation $x \log_{10} x = 1.2$ by Regula-false method correct to three decimal places. (7M+7M)
 b) Find the positive root of the equation $3x = \cos x + 1$ by iteration method.

UNIT - II

3. a) Using Gauss forward difference formula, Find Y (8), from the following table (7M+7M)

X	0	5	10	15	20	25
Y	7	11	14	18	24	32

- b) Find the interpolating polynomial $f(x)$ for the data of the following table

x	0	1	4	5
f(x)	4	3	24	39

(OR)

4. a) Using Gauss backward formula, find $f(42)$, from the following table (7M+7M)

X	20	25	30	35	40	45
f(x)	354	332	291	260	231	204

- b) Using Lagrange's interpolation formula find Y(10) from the following table

x	5	6	9	11
Y	12	13	14	16

UNIT - III

5. a) Evaluate $\int_0^2 \frac{dx}{x^3 + x + 1}$ by using Simpsons $1/3^{\text{rd}}$ rule with $h = 0.25$ (7M+7M)
 b) Evaluate $y(0.8)$ using Runge Kutta method given $y' = (x + y)^{\frac{1}{2}}$, $y(0.4) = 0.41$

(OR)

6. a) A rocket is launched from the ground. Its acceleration $a(t)$ measured every 5 seconds is tabulated below. Use trapezoidal rule to find the velocity and the position of the rocket at $t = 40$ seconds. (7M+7M)

t	0	5	10	15	20	25	30	35	40
a(t)	40.0	45.25	48.50	51.25	54.35	59.48	61.5	64.3	68.7

- b) Given $y' = x + \sin y$, $y(0) = 1$, compute $y(0.2)$ and $y(0.4)$ with $h = 0.2$ using modified Euler's method.

UNIT – IV

7. a) Find the Fourier series to represent $f(x) = x - x^2$ from $x = -\pi$ to $x = \pi$. (7M+7M)
 b) Obtain the sine series for $f(x) = x$ in $0 \leq x \leq \pi$.

(OR)

8. a) Obtain the Fourier series for the function $f(x) = \begin{cases} \pi x, & 0 \leq x < 1 \\ \pi(2 - x), & 1 \leq x \leq 2 \end{cases}$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$. (7M+7M)
 b) Find half range cosine series for $f(x) = x(2 - x)$ in $0 < x < 2$.

UNIT – V

9. a) Find the Fourier Transform of $\frac{1}{\sqrt{|x|}}$. (7M+7M)

- b) Find the Fourier integral representation for $f(x) = \begin{cases} 1 - x^2, & \text{for } |x| \leq 1 \\ 0, & \text{for } |x| > 1 \end{cases}$

(OR)

10. a) Find the inverse Fourier transform $f(x)$ of $F_s(p) = \frac{p}{1+p^2}$. (7M+7M)

- b) Find the Fourier cosine transform of e^{-ax} . Hence evaluate $\int_0^\infty \frac{\cos \lambda x}{x^2 + a^2} dx$

[B17 BS 1203]
I B. Tech II Semester (R 17) Regular Examinations
MATHEMATICS-III
(Common to all Branches)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

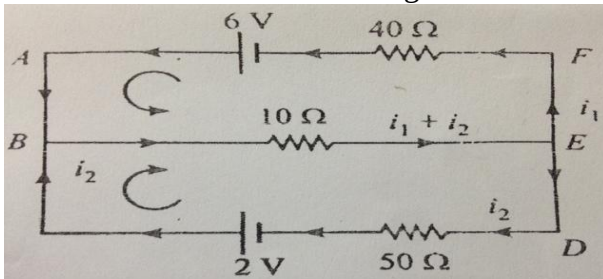
Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT - I

- 1.a) Solve the system of equations $20x + y - 2z = 17$, $3x + 20y - z = -18$,
 $2x - 3y + 20z = 25$ by Gauss –Siedel method. (7M+7M)

- b) Find the currents in the following circuit.



(OR)

2. a) Solve the system of equations $10x + y + z = 12$, $2x + 10y + z = 13$, $2x + 2y + 10z = 14$ by Gauss-elimination method. (7M+7M)

- b) Define rank and find the rank of the matrix A by reducing it in to its normal form where

A is:
$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}.$$

UNIT - II

3. a) Verify Cayley-Hamilton theorem and find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}. \quad (7M+7M)$$

- b) Reduce the quadratic form $2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ to canonical form by orthogonal transformation and hence find rank, index, signature and nature of the quadratic form.

(OR)

4. a) Find the eigen values and the corresponding eigen vectors of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}. \quad (7M+7M)$$

- b) If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, use Cayley-Hamilton theorem to find the value of $2A^5 - 3A^4 + A^2 - 4I$. Also find the inverse of A.

UNIT - III

5. a) Evaluate $\int_0^a \int_{\frac{x^2}{a}}^{2a-x} xy^2 \, dy dx$ by changing the order of integration. (7M+7M)

b) Establish the relation between Beta and Gamma functions.

(OR)

6. a) Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx \, dy$ by changing in to polar coordinates and hence deduce

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}. \quad (7M+7M)$$

b) Express $\int_0^1 x^m (1-x^n)^p dx$ in terms of Γ functions and hence evaluate

$$\int_0^1 x^5 (1-x^3)^{10} dx.$$

UNIT - IV

7. a) Find the directional derivative of $\phi(x, y, z) = x^2 yz + 4xz^2$ at the point (1, -2, -1) in the direction of the normal to the surface $f(x, y, z) = x \log z - y^2$ at (-1, 2, 1). (7M+7M)

b) Prove that $\text{div}(\text{grad } r^n) = n(n+1) r^{n-2}$ and $\text{curl}(\text{grad } \phi) = 0$ for any scalar function ϕ .

(OR)

8. a) Show that the vector field $\vec{F} = (x^2 + xy^2)\vec{i} + (y^2 + x^2y)\vec{j}$ is conservative and find the scalar potential function. (7M+7M)

b) Find the constants a, b such that the surfaces $5x^2 - 2yz - 9x = 0$ and $ax^2y + bz^3 = 4$ cut orthogonally at (1, -1, 2).

UNIT - V

9. a) Evaluate by Green's theorem $\oint_C [(y - \sin x)dx + \cos x \, dy]$ where C is the triangle enclosed by the lines $y=0$, $x=\pi/2$, $y=2x/\pi$. (7M+7M)

b) State Gauss Divergence theorem and use it to evaluate $\iint_S \vec{u} \cdot \vec{n} \, ds$ where $\vec{u} = x\vec{i} + y\vec{j} + z\vec{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = 9$.

(OR)

10. a) State Green's theorem in a plane and apply the theorem to evaluate

$$\oint_C (x^2y \, dx + y^3 \, dy) \text{ where C is the closed path formed by } y=x \text{ and } y=x^3 \text{ from } (0,0) \text{ to } (1,1). \quad (7M+7M)$$

b) Evaluate by Stokes' theorem $\oint_C [(x+y)dx + (2x-z)dy + (y+z)dz]$ where C is the boundary of the triangle with vertices (0,0,0), (1,0,0) and (1,1,0)

[B17 BS 1204]
I B. Tech II Semester (R 17) Regular Examinations
ENGINEERING PHYSICS
(Common to CIV,EEE & ME)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT – I

1. (a) Describe the interference phenomena in thin transparent films for reflected light and obtain the conditions for maxima and minima. [7M]
(b) Discuss the Fraunhofer diffraction of monochromatic light at a single slit. [7M]

(OR)

2. (a) Describe, with a neat sketch, the design and working of Michelson's interferometer. [7M]
(b) Explain how the resolving power of a grating can be determined. [7M]

UNIT – II

3. (a) Differentiate spontaneous and stimulated emission processes and obtain the Einstein's relation for spontaneous to stimulated emission coefficients. [7M]
(b) Define numerical aperture of an optical fiber and derive an expression for the same. [7M]

(OR)

4. (a) With neat sketches, explain the principle and working of He – Ne gas laser system. [7M]
(b) Explain the characteristics of lasers and mention the applications of lasers. [7M]

UNIT – III

5. (a) Discuss the electric fields induced due to time varying magnetic fields and deduce the Faraday's law. [7M]
(b) Describe any one method of detecting ultrasonics and mention the important applications of Ultrasonics. [7M]

(OR)

6. (a) Explain the concept of displacement current, and describe the significance of Maxwell's equations. [7M]
(b) What is magnetostriction and describe the magnetostriction method of producing Ultrasonics. [7M]

UNIT – IV

7. (a) What are matter waves and describe an experiment confirming the wave nature of electrons. [7M]
(b) What are the salient features of Kronig - Penny model. [7M]
8. (a) Explaining the physical significance of wave function of a particle derive the Schrodinger's time independent wave equation. [7M]
(b) Using band theory of solids how do you classify the materials. [7M]

UNIT - V

9. (a) What is a unit cell and describe the different crystal systems possible in solids. [7M]
(b) What are nano materials and explain the chemical vapour deposition method of fabricating nano materials. [7M]
- (OR)
10. (a) Define packing fraction and deduce the packing fraction for a simple cubic structure. [7M]
(b) Define the basic approaches of fabricating nano materials and discuss the sol-gel method. [7M]

[B17 BS 1204]

[B17 BS 1205]
I B. Tech II Semester (R 17) Regular Examinations
ENGINEERING CHEMISTRY
(Common to CSE,ECE & IT)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT - I

1. (a) Explain the mechanism of free radical Polymerization reaction with a suitable example. [7M]
(b) Distinguish between thermoplastic and thermosetting resins. [7M]

(OR)

2. (a) What are conducting Polymers? Discuss the applications of conducting Polymers. [7M]
(b) Write notes on Bu Na – S and Bu Na – N. [7M]

UNIT - II

3. (a) Explain the Proximate analysis of coal and give its significance. [7M]
(b) Explain the fractional distillation of crude oil. [7M]

(OR)

4. (a) Write notes on (i) Knocking (ii) CNG [7M]
(b) How Synthetic Petrol can be prepared by Bergius Process. [7M]

UNIT - III

5. (a) Explain the mechanism of electrochemical theory of corrosion with neat diagram. [7M]
(b) Describe briefly about cathodic Protection. [7M]

(OR)

6. (a) Explain Hydrogen – Oxygen fuel cell with neat cell diagram [7M]
(b) Discuss on various constituents of Paint. [7M]

UNIT - IV

7. (a) What is hardness? How it is determined by EDTA method? Explain. [7M]
(b) Describe with equations how water can be softened using Lime & Soda Process [7M]

(OR)

8. (a) Discuss various sterilizing methods used in municipal water treatment. [7M]
(b) Illustrate the reverse osmosis Process with a neat diagram. [7M]

UNIT - V

9. (a) Discuss chemistry involved in setting and hardening of cement? [7M]
(b) What are refractories? Discuss the classification of refractories. [7M]

(OR)

10. (a) Write the engineering applications of Liquid Crystals. [7M]
(b) Explain the stoichiometric defects in crystals. [7M]

[B17 BS 1205]

[B17 CS 1201]
I B. Tech II Semester (R 17) Regular Examinations
COMPUTER PROGRAMMING USING C
(Common to CIV,EEE & ME)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.
All questions carry equal marks.

UNIT-I

1. a) Discuss about computer languages. [7M]
- b) Explain c tokens. [7M]

(OR)

2. a) Explain different c operators. [7M]
- b) write about algorithm, pseudo code and flowchart. [7M]

UNIT-II

3. a) Discuss various looping techniques in c. [7M]
- b) Write a c program for summation of n numbers. [7M]

(OR)

4. a) Explain 2-D arrays and character arrays in c. [7M]
- b) Write a c program to find frequency of characters of a string. [7M]

UNIT-III

5. a) Explain parameter passing techniques in c. [7M]
- b) Write a c program for towers of Hanoi using recursive function. [7M]

(OR)

6. a) Explain storage classes in c. [7M]
- b) Write a c program for Fibonacci series using recursive function. [7M]

UNIT-IV

7. a) What is a pointer? How pointer variables are initialized. [7M]
- b) Write a program to print command line arguments on the screen. [7M]

(OR)

8. a) Discuss character pointers with examples. [7M]
- b) Write a c program to pass pointer variables as function arguments. [7M]

UNIT-V

9. a) Explain the difference between structure and union and write a program to find sum of marks in 3 subjects for a student using structures. [7M]
- b) Explain different bit-wise operators in c. [7M]

(or)

10. a) Explain about the input and output operations of a file. [7M]
- b) Write a c program to open a file and to print its contents on screen. [7M]

[B17 CS 1201]

[B17 CS 1202]
I B. Tech II Semester (R 17) Regular Examinations
OBJECT ORIENTED PROGRAMMING
THROUGH C++
(COMPUTER SCIENCE & ENGINEERING)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. A. What are the features of object programming language? [7M]
B. List the drawbacks of conventional programming? [7M]

(OR)

2. A. Explain array of objects with a suitable program? [7M]
B. Explain inline function with an example? [7M]

UNIT-II

3. A. Explain constructor overloading with an example? [7M]
B. Explain assignment overloading with a suitable example? [7M]

(OR)

4. A. Explain Dynamic initialization of Objects? [7M]
B. What is operator overloading? Write a C++ program illustrating overloading binary operators? [7M]

UNIT-III

5. A. Explain the concepts of pointers to objects? [7M]
B. What is virtual base class? Write a C++ program illustrating virtual base classes? [7M]

(OR)

6. A. Explain virtual function with an example? [7M]
B. Explain hybrid inheritance with an example? [7M]

UNIT-IV

7. A. What is an Exception? Explain about try, throw and catch with example? [7M]
B. Explain unformatted I/O operations with examples? [7M]

(OR)

8. A. Explain the principles of exception handling? [7M]
B. What are the String Characteristics? [7M]

UNIT-V

9. A. Explain about different types of containers? [7M]
B. Write a program for bubble sort using function templates? [7M]
- (OR)**
10. A. Explain the concepts of command line arguments. [7M]
B. Explain differences between templates and macros? [7M]

[B17 CS 1202]

[B17 CS 1203]

I B. Tech II Semester (R 17) Regular Examinations
DATA STRUCTURES
(Electronics and Communication Engineering)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.
All questions carry equal marks.

UNIT-I

1. A) Define data structure. Discuss different types of data structure their implementations applications. (7M)
B) Implement binary search technique using recursion. (7M)
- (OR)**
2. A) What is an array? Discuss different types of array with examples. (7M)
B) Rearrange following numbers using quick sort: (7M)
10, 6, 3, 7, 17, 26, 56, 32, 72

UNIT-II

3. A) Write an algorithm for basic operations of stack. (7M)
B) Explain the procedure to evaluate postfix expression. Evaluate the following postfix expression $7\ 3\ 4\ +\ -\ 2\ 4\ 5\ /\ +\ * \ 6\ /\ 7\ +\ ?$ (7M)
- (OR)**
4. A) Define Queue. Explain the operations of queue using arrays. (7M)
B) Explain the advantages of circular queue (7M)

UNIT-III

5. A) Define pointer. Explain Dynamically allocated storage using pointers. (7M)
B) Write an Algorithm for the operations of Linked stack (7M)
- (OR)**
6. A) Write an Algorithm for the operations of single Linked list (7M)
B) Explain polynomial addition using Linked List (7M)

UNIT-IV

7. A) What is a Binary tree. Explain threaded binary tree. (7M)
B) Explain Binary tree traversal techniques. (7M)
- (OR)**
8. A) Explain the operations of Binary search trees. (7M)
B) Define Max Heap. Write an algorithm for deletion of elements from Max Heap. (7M)

UNIT-V

9. A) What is a graph? Explain the properties of graphs. (7M)
B) Write breadth first traversal algorithm. Explain with an example. (7M)
- (OR)**
10. A) Define Minimum spanning tree. Explain Kruskal's Algorithm. (7M)
B) Write an Algorithm to find shortest path in a Graph (7M)

[B17 CS 1203]

[B17 ME 1201]

I B. Tech II Semester (R 17) Regular Examinations
ENGINEERING DRAWING
(Common to CSE,ECE & IT)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

Assume the missing data if any, suitably

UNIT-I

1. An inelastic string 145 mm long has its one end attached to the circumference of a circular disc of 40 mm diameter. Draw the curve traced out by the other end of the string, when it is completely wound around the disc, keeping the string always tight. [14 M]

[OR]

2. Two fixed points A and B are 100mm apart, Trace the complete path of a point P moving (in the same plane as that of A and B) in such a way that the sum of its distance from A and B is always the same and equal to 125mm. Name the curve and draw another curve parallel to and 25mm away from this curve. [14 M]

UNIT-II

3. A line AB, of 80 mm long has its end A, 15 mm in front of VP and 20 mm above HP. The other end B is 40 mm above HP and 50 mm in front of VP. Draw the projections of the line and determine the inclinations of the line with HP and VP. [14 M]

[OR]

4. (a) Draw the projections of a 75mm long straight line in the following positions: (i) parallel to and 30mm above the HP and in the VP; (ii) perpendicular to the VP, 25mm above the HP and its one end in the VP; (iii) Inclined at 30° to the HP and its one end 20mm above it, parallel to and 30mm in front of the VP. [7 M]

(b) Draw the projections of the following points on the same ground line, keeping the projectors 25mm apart. (i) Point A in the HP and lying 20mm behind the VP; (ii) Point B is 40mm above the HP and 25mm in front of the VP; (iii) Point C is 25mm below the HP and 25mm behind the VP; (iv) Point D is 15mm above the HP and 50mm behind the VP. [7 M]

UNIT-III

5. Draw a rhombus of diagonals 100 mm and 60 mm long, with the longer diagonal horizontal. The figure is the top view of a square of 100mm long diagonals, with a corner on the ground. Draw its front view and determine the angle which its surface makes with the ground. [14 M]

[OR]

6. A semicircular plate of 40mm diameter has its straight edge in the VP and inclined at 45° to the HP, the surface of the plate makes an angle of 30° with the VP. Draw its projections. [14 M]

UNIT-IV

7. A hexagonal pyramid, base 25mm side and axis 50mm long, has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the VP. Draw its projections. [14 M]

[OR]

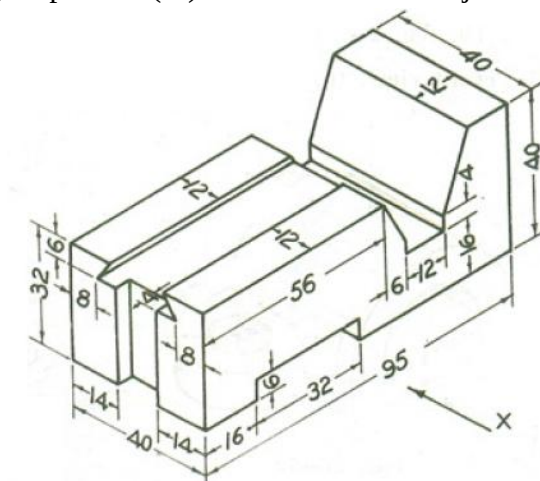
8. Draw the projections of a cylinder 75mm diameter and 100mm long, lying on the ground with its axis inclined at 30° to the VP and parallel to the ground. [14 M]

UNIT-V

9. A square pyramid with base side 40mm and height 60mm is resting on a cube of sides 50mm, the axes of the cube and the pyramid being in the same line. Two sides of the base of the pyramid are parallel to the edges of the cube. Draw the isometric view. [14 M]

[OR]

10. Draw (i) Front View (ii) Top View (iii) Side View of the object shown below: [14 M]



All the dimensions are in mm

[B17 ME 1201]

[B17 CE 1201]

I B. Tech II Semester (R 17) Regular Examinations

BUILDING MATERIALS AND CONSTRUCTION
(For Civil)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. a) Explain the classification of bricks. (Including IS classification) [7 M]
b) Explain the following clay products: [7 M]
 - i) Stoneware
 - ii) Terra-cotta

(OR)

2. a) Explain the term “Quarrying of stones”? [7 M]
b) Classify tiles and explain them with neat sketches. [7 M]

UNIT-II

3. a) What is seasoning of timber? Explain the defects due to seasoning of timber. [7 M]
b) Explain the following wood based products: [7 M]
 - i) Block Boards
 - ii) Particle Boards

(OR)

4. a) List various classifications of plywood. [7 M]
b) Draw the cross-section of a tree and explain the various details. [7 M]

UNIT-III

5. a) Define Specific gravity, Bulk density and Porosity of aggregates. [7 M]
b) Explain the manufacturing process of cement by “Dry” process? [7 M]
- (OR)**
6. a) Explain the term “Bulking of sand”? [7 M]
b) State and explain various laboratory tests for testing OPC? [7 M]

UNIT-IV

7. a) What are FAL-G blocks and Concrete blocks [7 M]
b) What are the characteristics of an ideal paint? [7 M]

(OR)

8. a) Explain various closers in Brick masonry with neat sketches? [7 M]
b) What is a foundation? Explain different types of foundations? [7 M]

UNIT-V

9. a) What is Roofing? Explain Madras terrace Roof? [7 M]
b) Define Form work and explain the different types of form work. [7 M]

(OR)

10. a) What is Scaffolding? Explain the different types of Scaffoldings? [7 M]
b) List out various staircases and explain any two them with neat sketches. [7 M]

[B17 CE 1201]

[B17 EC 1201]

I B. Tech II Semester (R 17) Regular Examinations

ELEMENTS OF ELECTRONICS ENGINEERING
(Common to CSE & IT)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT I

1. a) Explain in detail about drift and diffusion currents. (7M)
- b) Explain Hall Effect and its applications in detail. (7M)

OR

2. a) What are the differences between Avalanche breakdown and Zener Breakdown. (7M)
- b) Explain the basic operation of semiconductor diode with v-I characteristics. (7M)

UNIT II

3. a) Explain the V-I Characteristic of Zener Diode, and state its applications. (7M)
- b) Explain Tunneling phenomenon and V-I Characteristics of Tunnel diode. (7M)

OR

4. a) Derive expression for the ripple factor and efficiency of half wave rectifier without filter. (7M)
- b) With neat diagram, explain the operation of full wave rectifier and obtain expression for with filter Ripple factor. (7M)

UNIT III

5. a) Plot the input and output characteristics of transistor in CE configuration and explain the shape of the characteristics. (7M)
- b) What is early effect and what are its consequences. (7M)

OR

6. a) Plot the input and output characteristics of the transistor in CB configuration and explain shape of the curves. (7M)
- b) Define α and β . Derive the relation between α and β . (7M)

UNIT IV

7. a) Draw and explain the Self biasing circuit. Derive an expression for Stability factor S. (7M)
- b) Explain the phenomenon of Thermal runaway. (7M)

OR

8. a) Explain any two types of bias compensation. (7M)
- b) What are the reasons for the instability of operating point? Briefly explain the methods of stabilization of operating point. (7M)

UNIT V

9. a) Explain the operation of JFET with Drain and Transfer characteristics. (7M)
- b) What are the differences between JFET and BJT? (7M)

OR

10. a) Explain the operation of Enhancement MOSFET in details. (7M)
- b) Define g_m , r_d and μ of JFET and give the relation between them. (7M)

[B17 EC 1201]

[B17 EE 1201]

I B. Tech II Semester (R 17) Regular Examinations

CIRCUIT THEORY
(Electrical Electronics Engineering)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

UNIT-I

1. a) Obtain the expressions for star-delta and delta-star equivalence of resistive network. (7M)
- b) Find the value of resistance R, if the current is $I=11$ A and source voltage is 66 V as shown in figure. (7M)

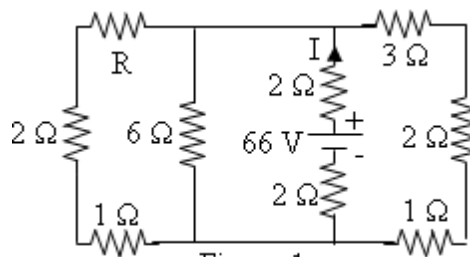


Figure:1

(OR)

2. a) Explain Source Transformation with suitable examples. (7M)
- b) Use the nodal analysis to determine voltage at node 1 and the power supplied by the dependent current source in the network shown in figure. (7M)

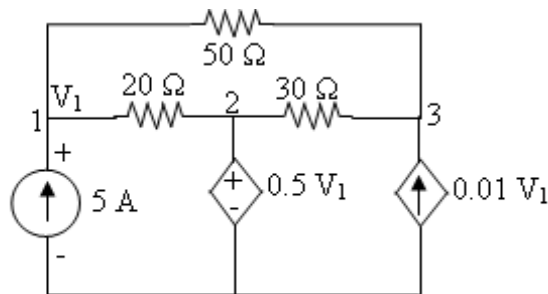


Figure:2

UNIT-II

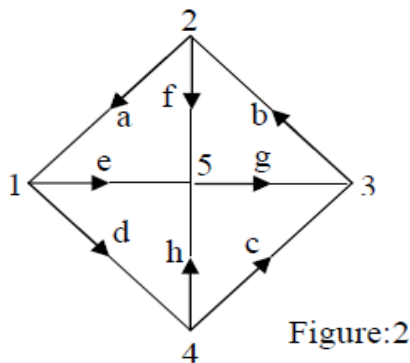
- 3.a) Explain the procedure for obtaining fundamental tie-set matrix of given network. (7M)
- b) Draw the oriented graph of a network with fundamental cut-set matrix as shown below. (7M)

Twigs				Links		
1	2	3	4	5	6	7
1	0	0	0	-1	0	0
0	1	0	0	1	0	1
0	0	1	0	0	1	1
0	0	0	1	0	1	0

Also find number of cut-sets and draw them.

(OR)

- 4.a) For the network graph shown in figure, draw all possible trees. For any one of these trees, prepare a cut-set schedule and obtain the relation between tree-branch voltages and branch voltages. (7M)



- b) Describe the procedure to construct the dual of a network with an example. (7M)

UNIT-III

- 5.a) A ring has a mean diameter of 21 cm and cross sectional area of 10 cm^2 . The ring is made up of semi-circular sections of cast iron and cast steel with each joint having reluctance equal to an air gap of 0.2 mm. Find the ampere turns required to produce a flux of 0.8 milli Wb. The relative permeability of cast steel and cast iron are 800 & 166 respectively. Neglect fringing and leakage effects. (7M)
- b) Two identical coupled coils have an equivalent inductance of 80 mH when connected series aiding and 35 mH in series opposing. Find L_1 , L_2 , M and K . (7M)

(OR)

6. a) Derive the relationship between Flux, MMF and Reluctance. (7M)
- b) A coil is wound uniformly with 400 turns over an iron ring having a mean circumference of 50 cm and a cross section of 0.4 cm^2 . If the coil has resistance of 10Ω and is connected across a 50V DC supply, calculate the m.m.f of the coil, magnetic field strength, magnetic field density, total flux and reluctance of the ring. (7M)

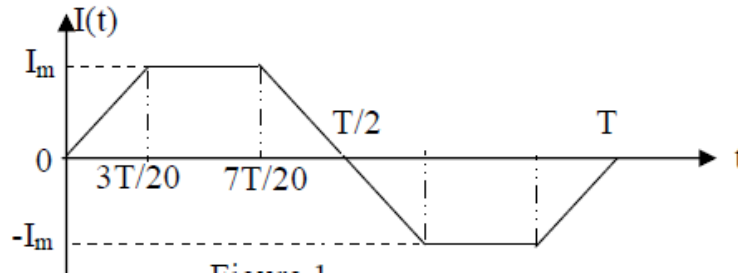
UNIT-IV

7. a) Define the following: (7M)
- Amplitude of an alternating quantity
 - Instantaneous value of an alternating quantity
 - Frequency

b) Show that power consumed in a purely inductive circuit is zero when sinusoidal voltage is applied across it. (7M)

(OR)

8. a) Find the average value, r.m.s value, form factor and peak factor for the wave form shown in figure. (7M)



b) A coil of inductance L and resistance R in series with a capacitor is supplied at a constant voltage from a variable frequency source. If the frequency is ω_r , find in terms of L , R and ω_r the values of those frequencies at which the circuit current would be half as much as that at resonance. Hence or otherwise determine the bandwidth and selectivity of the circuit. (7M)

UNIT-V

9. a) Explain the relationship between line and Phase quantities in delta connected circuits? (7M)

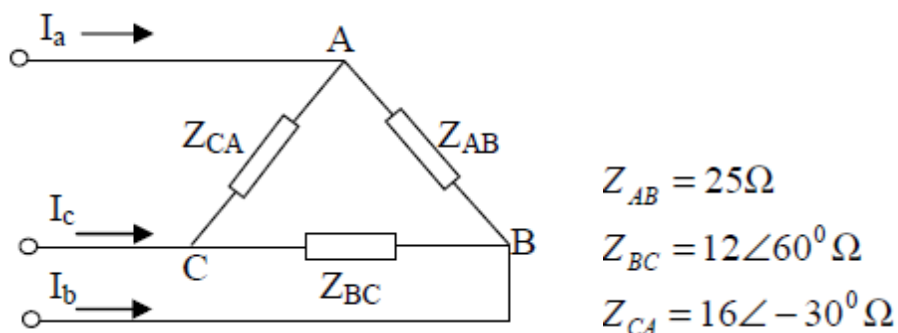
b) A balanced star connected load of $(4+j3) \Omega$ per phase is connected to a balanced 3ϕ 400V supply.

The phase current is 12 A. Find a) active power b) reactive power c) Apparent power (7M)

(OR)

10. a) A four-wire star-star circuit has $V_{an} = 120 \angle 120^\circ$, $V_{bn} = 120 \angle 0^\circ$, $V_{cn} = 120 \angle -120^\circ$ V. If the impedances are $Z_{an} = 20 \angle 60^\circ$, $Z_{bn} = 30 \angle 0^\circ$ and $Z_{cn} = 40 \angle 30^\circ \Omega$, find the current in the neutra line. (7M)

b) For the circuit shown in figure 3, the line voltage is 240 V. Take V_{ab} as reference and determine following: i) phase currents, ii) line currents, iii) total power absorbed in the load. Also draw Phasor diagram (7M)



[B17 EE 1201]

[B17 EE 1202]
I B. Tech II Semester (R 17) Regular Examinations
BASIC ELECTRICAL & ELECTRONICS ENGINEERING
(Mechanical Engineering)
MODEL QUESTION PAPER

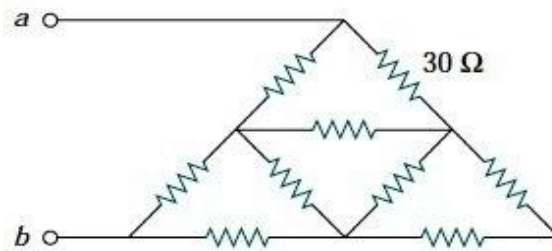
Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.
All questions carry equal marks.

UNIT-I

1. (a) State and explain Kirchhoff's Laws with example. [7M]
(b) Find the equivalent resistance R_{ab} for the circuit shown below. All the resistor values are 30Ω . [7M]



OR

2. (a) Define Dynamically Induced E.M.F and derive expression for it. [7M]
(b) A coil having an inductance 60mH is carrying a current of 60A . Calculate the Self-induced EMF in the coil. When the current in the coil reversed in 30milliseconds . [7M]

UNIT-II

3. (a) Derive the EMF equation of DC generator [7M]
(b) A shunt generator supplies a load of 7.5KW at 200V , Calculate the generated emf if armature resistance is 0.6Ω and field resistance of 80Ω . [7M]

OR

4. (a) Derive the torque equation of the DC motor. [7M]
(b) An 8-pole, wave-connected armature has 600 conductors and is driven at 625 rev/min . If the flux per pole is 20 mWb , determine the generated E.M.F. [7M]

UNIT-III

5. (a) Derive the EMF equation of a single phase transformer. [7M]
(b) A 200 KVA rated transformer has a full-load copper loss of 1.5 kW and an iron loss of 1 kW. Determine the transformer efficiency at full load & half load for 0.85 power factor. [7M]

OR

6. (a) Explain the operation of Transformer under NO-LOAD with phasor diagram. [7M]
(b) An ideal 25KVA Transformer has 500 turns on primary and 40 turns on the secondary winding. The primary winding is connected to 3000 V, 50Hz supply. Calculate (i) Primary and secondary currents (ii) Secondary EMF (iii) Maximum flux. [7M]

UNIT-IV

7. (a) Draw and explain the slip-Torque Characteristics of Three phase Induction motor. [7M]
(b) The frequency of the supply to the stator of a 6-pole induction motor is 50 Hz and the rotor frequency is 2 Hz. Determine (i) the slip, and (ii) the rotor speed in r.p.m [7M]

OR

8. (a) Derive the EMF equation of Alternator [7M]
(b) Obtain the Voltage Regulation of Alternator by synchronous impedance method [7M]

UNIT-V

9. a) Explain the operation of Diode in Forward and reverse bias conditions and draw V-I characteristics. [7M]
(b) Explain the operation of Zener diode and draw its V-I characteristics [7M]

OR

10. (a) Draw the circuit diagram of Bridge rectifier and explain its operation. [7M]
(b) Explain how the transistor acts as an amplifier. [7M]

[B17 EE 1203]
I B. Tech II Semester (R 17) Regular Examinations
ELEMENTS OF ELECTRICAL ENGINEERING
(Electronics and Communication Engineering)
MODEL QUESTION PAPER

Time: 3 hours

Max. Marks: 70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a) State and explain Kirchhoff's Laws with example. [7M]
- (b) Derive star-delta and delta- star Transformation for Equal resistances [7M]

(OR)

2. (a) Define Dynamically Induced E.M.F and derive expression for it. [7 M]
- (b) A coil having an inductance 60mH is carrying a current of 60A. Calculate the Self-induced EMF in the coil. When the current in the coil reversed in 30milliseconds. [7M]

UNIT-II

3. (a) Derive the EMF equation of DC generator [7M]
- (b) A series motor drives a load at 1500 r.p.m and takes a current of 20A when the supply voltage is 250V if the total resistance of the motor is 1.5 ohms and the iron, friction and windage losses amount to 400W. Determine the efficiency of the motor. [7M]

(OR)

4. (a) Derive the Torque equation of DC motor. [7M]
- (b) A shunt generator supplies a load of 7.5KW at 200V, Calculate the generated emf if armature resistance is 0.6Ω and field resistance of 80Ω . [7M]

UNIT-III

5. (a) Explain the operation of Transformer under NO-LOAD with phasor diagram. [7M]
- (b) An ideal 25KVA Transformer has 500 turns on primary and 40 turns on the secondary winding. The primary winding is connected to 3000 V, 50Hz supply. Calculate (i) Primary and secondary currents (ii) Secondary EMF (iii) Maximum flux. [7M]

(OR)

6. (a) Derive the EMF equation of a single phase transformer. [7M]

(b) A 25-kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000-V, 50-Hz supply. Find the full-load primary and secondary currents, the secondary e.m.f. and the maximum flux in the core. Neglect leakage drops and no-load primary current [7M]

UNIT-IV

- 7 (a) Explain the Slip - Torque Characteristics of Three phase Induction Motor. [7M]
(b) A 3-Phase Induction Motor is Running at 5% slip. The Output is 36.75KW and Total Mechanical losses are 1.5KW. Estimate the copper losses in the rotor. If the stator losses are 4KW, estimate the efficiency of the Motor. [7M]

(OR)

8. (a) Define Slip and Rotor Frequency in Detail. [7M]
(b) The Power Input to 3- ϕ Induction motor is 55Kw. Total stator losses Equal to 2.2Kw. Find
(i) Rotor copper loss (ii) Mechanical Power developed if the motor is running at a speed of 720rpm at 50Hz supply with 4poles. [7M]

UNIT-V

9. (a) Derive the EMF equation of Alternator [7M]
(b) Obtain the Voltage Regulation of Alternator by SYNCHRONOUS IMPEDENCE METHOD. [7M]

(OR)

10. (a) Explain the operation of PMMC with neat sketches [7M]
(b) Explain Deflecting, controlling and damping Torques with neat sketches [7M]

[B17 EE 1203]

[B17 BS 2101]
II B. Tech I Semester (R 17) Regular Examinations
MATHEMATICS-IV
Common to CE,ECE,EEE & ME
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a). Prove that the function defined by 7M

$$f(z) = \frac{(1+i)x^3 - (1-i)y^3}{x^2 + y^2} \quad (z \neq 0) \text{ and } f(0) = 0 \text{ is continuous and the Cauchy-}$$

Riemann equations are satisfied at the origin, yet $f'(0)$ do not exist.

- (b). Find the complex potential w and velocity potential ϕ when its stream 7M

function is $\psi = \tan^{-1}\left(\frac{y}{x}\right)$

(OR)

2. (a). If $f(z)$ is a regular function of z , prove that 7M

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$$

- (b). Find the bi-linear transformation which maps the points $Z = 1, -1, \infty$ onto the 7M
points $w = 1+i, 1-i, 1$. Hence find (a) the critical points and (b) the invariant
points of this transformation.

UNIT-II

3. (a). Solve $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$, given that $u = 3e^{-y} - e^{-5y}$ when $x=0$ 7M

- (b). A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest 7M
in its equilibrium position. If it is vibrating by giving to each of its points a
velocity $\lambda x(l-x)$, find the displacement of the string at any distance x from
one end at any time t .

(OR)

4. (a). A homogeneous rod of conducting material of length 100 cm has its ends kept 7M
at zero temperature and the temperature initially is

$$u(x, 0) = \begin{cases} x, & 0 \leq x \leq 50 \\ 100 - x, & 50 \leq x \leq 100 \end{cases}$$

Find the temperature $u(x, t)$ at any time.

- (b). Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for $0 < x < \pi, 0 < y < \pi$ with conditions given $u(x, \pi) = 0,$ 7M
 $u(0, y) = 0, u(\pi, y) = 0, u(x, 0) = \sin^2 x.$

UNIT-III

5. (a). Form the difference equation generated by $y_n = (A + Bn)3^n$ 7M

- (b). Solve the difference equation $y_{n+2} + y_{n+1} - 56y_n = 2^n(n^2 - 3)$ 7M

(OR)

6. (a). If $Z(u_n) = \frac{z}{z-1} + \frac{z}{z^2+1}$ find the Z-transform of u_{n+2} 7M
- (b). Using Z-transforms, solve $u_{n+2} - 2u_{n+1} + u_n = 3n + 5$. 7M

UNIT-IV

7. (a). Twenty identical coins each with probability P of showing heads are tossed. The Probability of heads showing on 10 coins is same as that of heads showing on 11 coins. Find P . 7M
- (b). X is a normal variate with mean 30 and standard deviation 5. Find the probability that (i) $26 \leq X \leq 40$ (ii) $X \geq 45$ (iii) $|X - 30| > 5$ 7M

(OR)

8. (a). Derive moment generating function of Poisson distribution. 7M
- (b). In a Normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. 7M

UNIT-V

9. (a). A sample of height of 6400 soldiers has a mean of 67.85 inches and a standard deviation of 2.56 inches while a simple sample of heights of 1600 sailors has a mean of 68.55 inches and a standard deviation of 2.52 inches. Do the data indicate that the sailors are on the average taller than soldiers? 7M
- (b). A die is tossed 960 times and it falls with 5 upwards 184 times. Is the die biased? 7M

(OR)

10. (a). A sample of 10 measurements of the diameter of a sphere gave a mean of 12 cm and a Standard deviation of 0.15 cm. Find 95% confidence limits for the actual diameter. 7M
- (b). The number of accidents in a street during a particular week is given below: 7M
- | | | | | | | |
|-------------------|-----|-----|-----|-----|-----|-----|
| Days: | Mon | Tue | Wed | Thu | Fri | Sat |
| No. of accidents: | 5 | 4 | 3 | 7 | 6 | 5 |
- Test the hypothesis that the number of accidents does not depend on the day of the week.

[B17 BS 2101]

[B17 EC 2101]
II B. Tech I Semester (R 17) Regular Examinations
ELECTRONICS DEVICES AND CIRCUITS
Common to ECE & EEE
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

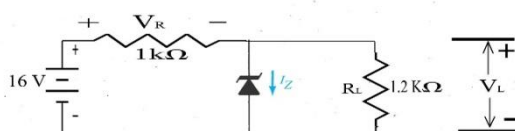
1. a). Distinguish between drift current and diffusion current in a semiconductor. Obtain expressions for drift current density and diffusion current density for both electrons and holes. 7M
- b). Discuss potential variation in a graded semiconductor and hence derive Mass Action law. 7M

(OR)

2. a). What is Hall effect and what are its applications? 6M
- b). Derive the continuity equations for both holes and electrons. 8M

UNIT-II

3. a). Discuss in detail the various current components in a pn-diode and hence obtain diode current equation. 7M
- b). For the circuit shown, $P_{zmax} = 30mW$, determine V_L , V_R , I_Z and P_Z 7M

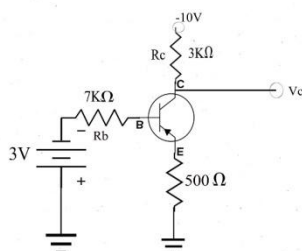


(OR)

4. a). With a neat diagram, explain the operation of a full wave rectifier and obtain expressions for ripple factor and efficiency. 7M
- b). A 220V,50Hz ac voltage is applied to the primary of a 4:1 step down transformer and this transformer is connected to a bridge rectifier with a load resistance $R_L = 1K\Omega$. Assuming ideal diodes, determine (i). V_{dc} . (ii). DC power delivered to load P_{dc} . (iii). PIV of each diode. (iv). I_{dc} . (v). Power rating of each diode. 7M

UNIT-III

5. a). Plot input and output characteristics of the transistor in CE configuration and explain the shape of the curves. 8M
- b). 6M



For the circuit shown , $\beta_{dc} = 100$ and $V_{BE} = 0.8V$.

- (a). find if the transistor is in cut off, saturation or active region.
- (b). Find V_C .
- (c). Find the minimum value of R_B for which the transistor operates in active region.

(OR)

6. a). Explain how self bias circuit improves stability of operating point and obtain an expression for the stability factor 'S' for the self bias circuit 7M
- b). Explain how diodes can be used to compensate against variations in V_{BE} and I_{CO} due to change in temperature. 7M

UNIT-IV

7. a). Compare JFET and BJT. 6M
- b). Explain the construction of JFET and explain the operation of JFET with drain and transfer characteristics. 8M

(OR)

8. a). Explain the construction of Enhancement MOSFET and explain the operation with Drain and Transfer characteristics. 7M
- b). Design a self-bias network using a JFET with $I_{DSS} = 8mA$ and $V_P = -6V$ to have a Q-Point at $I_{DQ} = 4mA$ with a supply voltage $V_{DD} = 14V$. Assume that $R_D = 3R_S$ and use standard values. 7M

UNIT-V

9. a). Derive expressions for current gain, Voltage gain, Input impedance and Output impedance of a transistor amplifier using h-Parameters. 8M
- b). Compare CB, CE and CC amplifier configurations. 6M

(OR)

10. a). Discuss the effects of Emitter resistance, bypass and Coupling capacitors in CE amplifier. 7M
- b). Obtain expressions for current gain, Voltage gain, Input impedance and Output impedance of emitter follower using h-Parameters. 7M

[B17 EC 2101]

[B17 EC 2102]
II B. Tech I Semester (R 17) Regular Examinations
SWITCHING THEORY AND LOGIC DESIGN
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a). Convert 1096_{10} in to Binary, Octal, Hexadecimal, BCD and Excess-3. **6M**
 (b). Simplify the following functions by using basic Boolean laws **8M**
 i). $\bar{x}\bar{z} + xyz + x\bar{z} = Y$
 ii). $\bar{A}\bar{B}C + (\bar{B}+\bar{C})(\bar{B}+\bar{D}) + \overline{(A+C+D)} = Y$

(OR)

2. (a). Convert 8421_{10} in to Binary, Octal, Hexadecimal, BCD and Excess-3. **6M**
 (b). i). Prove that $(A \oplus B) \oplus C = A \oplus (B \oplus C)$ **8M**
 ii). Simplify $\bar{A}B + ABD + A\bar{B}CD + BC = Y$

UNIT-II

3. (a). i. Find Standard Sum of Products form for $A + (\bar{A}\bar{B} + \bar{A}C) = Y$ **7M**
 ii. Find Standard Product of Sums for $A+C+\bar{B}\bar{D} = Y$
 (b). Simplify using K-Map $f(A,B,C,D) = \sum m(0,1,3,7,8,12) + \sum d(5,10,13,14)$ and draw the corresponding circuit with only NAND gates. **7M**

(OR)

4. (a). Simplify the given function by using K-map **7M**
 $f(A,B,C,D) = \sum m(0,1,4,5,7,11,15) + \sum d(10,13)$
 (b). Find Prime Implicants of $f(w,x,y,z) = \sum m(1,4,6,7,8,9,10,11,15)$ by using Quine - McCluskey method. **7M**

UNIT-III

5. (a). Design an 8:1 Multiplexer with only NAND Gates. **7M**
 (b). Implement the given functions by using a 3 to 8 Decoder and logic Gates. **7M**
 (i) $f_1(A,B,C) = \sum m(0,1,5,7)$
 (ii) $f_2(A,B,C) = \sum m(1,3,5,7)$

(OR)

6. (a). Design and draw a BCD to 7 segment Decoder circuit for active low outputs. **8M**
 (b). Design and draw Half adder and Full adder circuits. **6M**

UNIT-IV

7. (a). Convert D-FF in to JK-FF. **6M**
 (b). What is race around problem in a JK flip-flop and explain the method to avoid it by using Master Slave JK-FF. **8M**

(OR)

8. (a). Draw a 3-bit right shift register and explain the modes SIPO, SISO, PIPO and PISO. **7M**
 (b). Design Mod-5 Ring counter and draw its output waveforms. **7M**

UNIT-V

9. (a). Explain the terms race, critical race and non-critical race by taking an example. **6M**
(b). Design a Sequence detector which produces an output 0 every time the sequence 0101 is detected, an output 1 at all other times. Assume that overlapping sequence is allowed. Use D-FFs in your design. **8M**

(OR)

10. (a). Explain different types of Hazards. **6M**
(b). What is lockout in counters? Design a Mod-10 synchronous counter by using D-Flip- Flops. **8M**

[B17 EC 2102]

[B17 EC 2103]
II B. Tech I Semester (R 17) Regular Examinations
SIGNALS AND SYSTEMS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

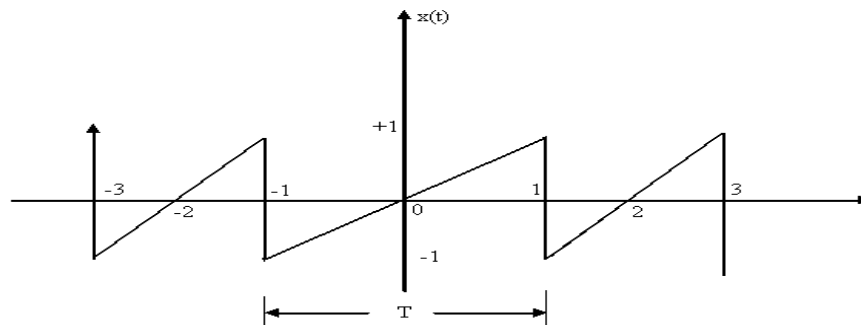
1. (a). Explain all classification of signals with examples for each category. 7M
- (b). Determine the power and RMS value of the following signals 7M
- (1). $x(t) = 5 \cos\left(50t + \frac{\pi}{3}\right)$ (2). $x(t) = 10 \cos 5t \cos 10t$

(OR)

2. (a). Determine whether the following systems are time invariant or not. 7M
- (1). $y(t) = tx(t)$ (2). $y(n) = x(2n)$
- (b). Prove the energy of the power signal is infinite over infinite time. 7M

UNIT-II

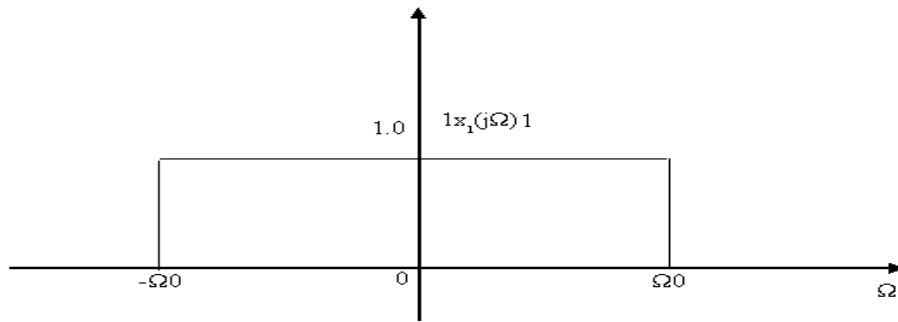
3. (a). Find the trigonometric Fourier series for the periodic signal $x(t)$ as shown in figure below 7M



- (b). State and prove parseval's relation of Fourier series. 7M
- (OR)**
4. (a). State and prove the time shifting and time scaling properties of DTFS. 7M
 - (b). Explain Dirichlet conditions on Convergence of Fourier Series. 7M

UNIT-III

5. (a). State and prove the differentiation in the frequency domain property of DTFT. 7M
 - (b). Find $x(t)$ from $X(j\omega) = 1/(1+j\omega)^2$ 7M
- (OR)**
6. (a). Find the Fourier transform of the following: 7M
 - (i) $e^{at}u(-t)$
 - (ii) $te^{-at}u(t)$
 - (b). Find the inverse Fourier transform of $X(j\Omega)$ for the spectra shown in figure below. 7M



UNIT-IV

7. (a). State and prove final value theorem of Laplace transform. 7M
 (b). Compute the initial and final values for $X(S) = (2S+5)/S(S+3)(S+4)^2$ 7M
 (OR)
 8. (a). Explain ROC properties of Laplace transforms. 7M
 (b). Determine the bilateral Laplace transform and ROC for the following signals 7M
 (i) $e^{-2t} u(t-2)$ (ii) $e^{-4|t|}$

UNIT-V

9. (a). Find the inverse z-transform of the following: 7M
 (i) $X(z) = 1 / (1 + 3z^{-1} + 2z^{-2})$ ROC : $|z| > 2$.
 (ii) $X(z) = (z + 1) / (z^2 + 5z + 4)$ ROC : $|z| < 2$

- (b). Solve the following difference equation for $y(n)$ using z-transform for the specified Initial condition 7M

$$y(n) - y(n-1) + \frac{1}{4} y(n-2) = x(n); n \geq 0$$

$$\text{where } x(n) = 2 \left(\frac{1}{8} \right)^n; y(-1) = 2 \text{ and } y(-2) = 4.$$

(OR)

10. (a). A signal $x(t) = \sin c(150\pi t)$ is sampled at a rate of (i) 100Hz (ii) 200 Hz (iii) 300Hz. For each of these cases, explain if you can recover the signal $x(t)$ from the sampled signal. 7M
 (b). Find the z-transform and ROC for the following: 7M
 (i) $x(n) = a^n u(n)$
 (ii) $x(n) = -b^n u(-n-1)$.

[B17 EE 2104]
II B. Tech I Semester (R 17) Regular Examinations
NETWORK ANALYSIS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

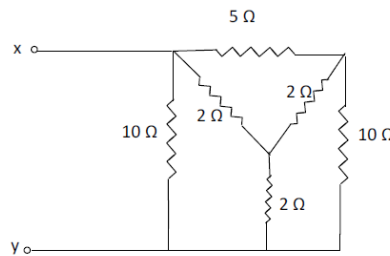
Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

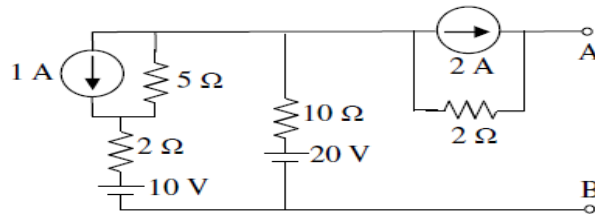
All questions carry equal marks.

UNIT-I

1. (a). Find the equivalent resistance between terminals x-y in the resistance network shown in figure below by using Y- Δ transformation. 7M

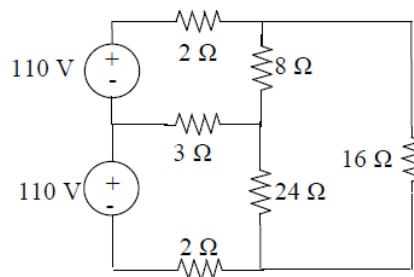


- (b). Find a single source equivalent at the terminals of a circuit shown in fig 7M

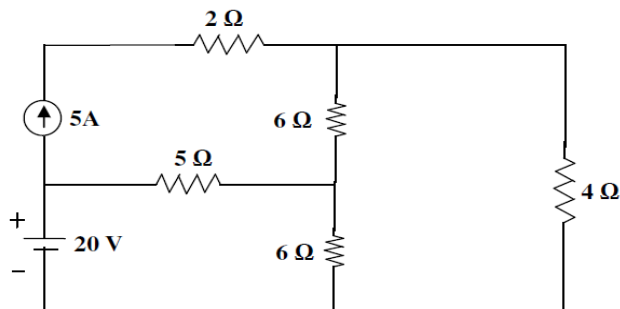


(OR)

2. (a). For the circuit shown in figure, find all the branch currents using nodal analysis. 7M

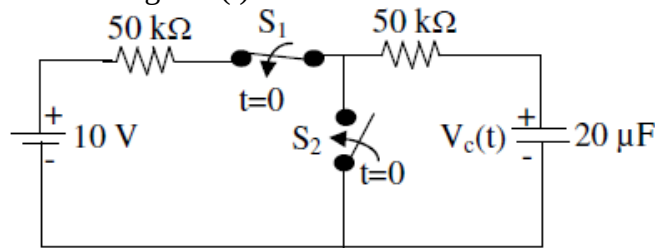


- (b). Find the current through 4 ohm resistor using mesh analysis in the figure shown below. 7M



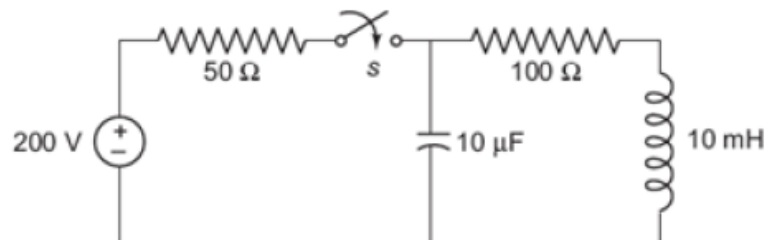
UNIT-II

3. (a). Derive an expression for the current in an RC circuit excited by a unit step voltage. 7M
- (b). In the circuit of figure, switch S_1 has been closed for a long time while switch S_2 has been open for a long time. At $t=0$, switch S_1 opens and switch S_2 closes. What is the value of voltage $V_c(t)$ for all $t>0$. 7M



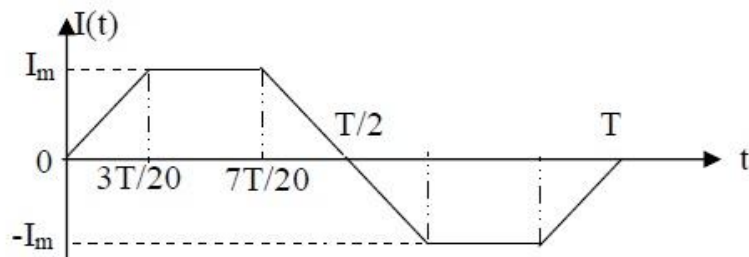
(OR)

4. (a). Derive an expression for the current in an RL circuit excited by a unit step voltage. 7M
- (b). When the switch is closed at $t = 0$, find the transient currents across inductor for the network shown in below Figure. Assume that initial current across the inductor is zero. 7M



UNIT-III

5. (a). A 200 V, 50 Hz AC supply is applied to a coil of 0.08 H inductance and 3.5 Ω Resistance connected in series with a 7.2 μF capacitor. Calculate (i) Impedance (ii) Current (iii) Phase angle between current and voltage (iv) power factor (v) power consumed. 7M
- (b). Find the average value, r.m.s value, form factor and peak factor for the wave form shown in figure. 7M

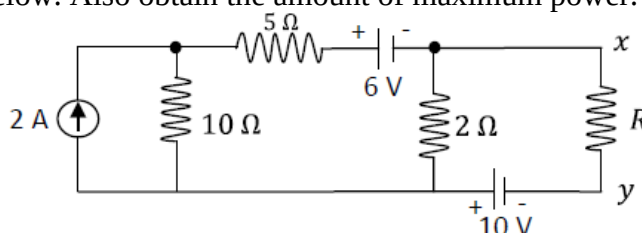


(OR)

6. (a). A coil having a resistance of 50 ohms and an inductance of 0.02 H is connected in parallel with a capacitor of 25 μF, across a 200 V, 50 Hz supply. Find the current in the coil and the capacitor. Also find the total current taken from the supply, the overall powerfactor and total power consumed. Draw the phasor diagram. 7M
- (b). A current of $(120-j50)$ A flows through a circuit when applied voltage is $(8+j12)$ V. Determine: (i) impedance (ii) power factor (iii) power consumed and reactive power 7M

UNIT-IV

7. (a). Find the value of R to have a maximum power transfer in the circuit shown below. Also obtain the amount of maximum power. 7M

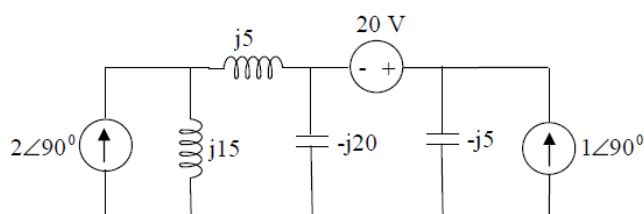


- (b). A series RLC circuit with $R = 10 \text{ ohms}$, $L = 0.4 \text{ H}$ and $C = 50 \text{ } \mu\text{F}$ has applied voltage of 200V with variable frequency. Calculate the resonant frequency, current at resonance, voltage across R , L and C . Also calculate the Q -factor, upper and lower half power frequencies and bandwidth. 7M

(OR)

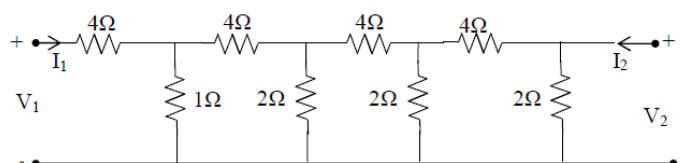
8. (a). Find the voltage across $-j20 \text{ ohms}$ capacitor using superposition theorem in below Figure All impedance values are in ohms. 7M

- (b). 7M



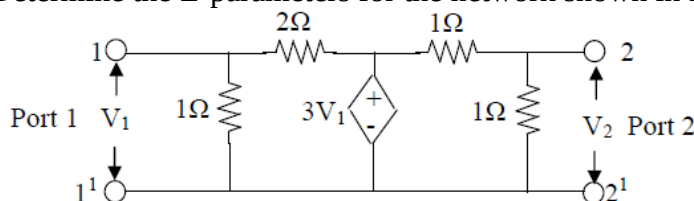
UNIT-V

9. (a). Express Y -parameters in terms of $ABCD$ and Z -parameters. 7M
 (b). Determine the $ABCD$ parameters of the network shown in figure. 7M

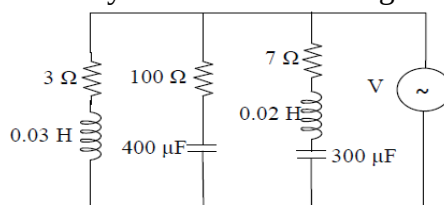


(OR)

10. (a). Determine the Z -parameters for the network shown in fig. 7M



- (b). Explain the concept of Duality. Obtain dual of the given network. 7M



[B17 EE 2104]

[B17 EC 2104]
II B. Tech I Semester (R 17) Regular Examinations
PROBABILITY THEORY & RANDOM PROCESSES
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a). State and prove Baye's theorem. 7M
- (b). A single card is drawn from a 52 card deck 7M
 - (i) What is the probability that the card is a jack?
 - (ii) What is the probability that the card will be a 5 (or) smaller?
 - (iii) What is the probability that the card is a red 10?

(OR)

2. (a). State and prove Addition theorem of probability. 7M
- (b). In a class room, 70% are above average, 20% are average and 10% are below average. Suppose that 20% of above average, 10% of average and 20% of below average students fail in a subject. What is the probability that a randomly selected student is an average student who failed? 7M

UNIT-II

3. (a). Define Distribution and Density functions. List the properties of them. Prove any one property of each. 7M
- (b). A random variable X has a probability density function 7M

$$f_X(x) = \frac{c}{x^2+1} \quad -\infty < x < \infty$$

Find the constant 'c' and also find the probability distribution function of the random variable X.

(OR)

4. (a). The joint probability density function of two random variables x and y is given by $(x, y) = e^{-(x+y)}$ If $x \geq 0, y \geq 0$, = 0 otherwise. Find the marginal density functions of x and y. Also prove that x and y are independent. 7M
- (b). A random variable X has a Gaussian density function with mean '0' and variance '1'. Find the probability density function of the random variable Y defined as $Y = X^2$. 7M

UNIT-III

5. (a). A random variable has a probability density function 7M

$$f_X(x) = \frac{5}{4}(1-x^4) \text{ for } 0 < x \leq 1, = 0 \text{ elsewhere}$$

Find (i) E(X) (ii) E(4X+2) (iii) E(X²).
 - (b). State and prove Chebyshev's inequality. 7M
- (OR)
6. (a). Compute the characteristic function for a random variable X with 7M

$$f_X(x) = \frac{1}{2}e^{-|x|}, \quad -\infty < x < \infty.$$
 - (b). State and Prove Central Limit Theorem. 7M

UNIT-IV

7. (a). Define Autocorrelation and power spectral density for a random process. List their Properties. 7M

- (b). Determine the PSD of a stationary process whose autocorrelation is given by $R_{XX}(\tau) = \sigma^2 e^{-\alpha|\tau|}$. 7M

(OR)

8. (a). Prove that the random process $X(t) = A \cos(\omega_0 t + \theta)$ is wide sense stationary where A, ω_0 are constants and θ is uniformly distributed random variable on the interval $(0, 2\pi)$. 7M

- (b). Distinguish between stationary and Ergodic random processes 7M

UNIT-V

9. (a). Find the input autocorrelation function, output autocorrelation function and output spectral density of RC low pass filter, when the filter is subjected to a white noise of spectral density $N_0/2$. 7M

- (b). Show that the Narrow band noise process can be expressed as In-phase and Quadrature components. 7M

(OR)

10. (a). What is Low pass and Band pass noise representation and write their properties? 7M

- (b). Write short notes on (i) Poisson Process (ii) Gaussian Process 7M

[B17 EC 2104]

[B17 EC 2201]
II B. Tech II Semester (R 17) Regular Examinations
ELECTRONICS CIRCUIT ANALYSIS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT - I

- 1 a) Derive expression for voltage gain and current gain of a Two stage RC coupled amplifier using low frequency model. 7M
- b) A transistor is connected as a CE amplifier with load resistance of $10K\Omega$. The parameters are $h_{ie}=5K\Omega$ and $h_{fe}=330$. Calculate the overall gain for mid frequency range when four such stages are connected in cascaded RC coupling. Assume $R_s=0$. 7M

OR

- 2 a) Explain what you understand by the term bandwidth shrinkage factor as applied to cascaded RC stages. Derive the expression for the same. 6M
- b) Derive the expression for short circuit current Gain-Bandwidth product of a CE amplifier. 8M

UNIT - II

- 3 a) Derive expressions for input impedance and output impedance in case of a voltage series feedback topology. 8M
- b) An amplifier gain changes by $\pm 10\%$ using negative feedback amplifier is to be modified to yield gain of 100 with 0.1% variation. Find required loop gain and amount of negative feedback. 6M

OR

- 4 a) Derive expressions for input impedance and output impedance for a current shunt feedback topology. 8M
- b) Determine voltage gain, input impedance and output impedance of an amplifier with feedback for voltage series feedback having $A= 100$, $R_i=10K\Omega$, $R_o=20 K\Omega$ and $\beta = 0.1$. 6M

UNIT - III

- 5 a) Derive an expression for the frequency of oscillations of an RC- Phase shift oscillator. 8M
- b) In a Colpitts oscillator $C_1 = 0.001 \mu F$ and $C_2 = 0.01 \mu F$ and $L = 5 \mu H$. Calculate
(i) Frequency of oscillations.
(ii) If 'L' is doubled, find the new frequency. 6M

OR

- 6 a) Describe Crystal oscillator. 6M
- b) Draw the circuit of a Hartley oscillator and explain how oscillations build up in the circuit. Derive the expressions for frequency of oscillations. 8M

UNIT - IV

- 7 a) Differentiate single tuned and stagger tuned amplifiers. 8M
- b) Derive expression for efficiency of class A power amplifier with Transformer coupled load. 6M

OR

- 8 a) Derive expression for the gain of a Single Tuned voltage amplifier. 7M
- b) Explain how even harmonic distortion is reduced in a Class-A push-pull power amplifier. 7M

UNIT - V

- 9 a) With the help of neat diagrams, explain the following applications using Op-Amp. 8M
- (i) Non Inverting amplifier.
- (ii) Summing amplifier.
- b) Explain the concept of “Virtual ground” for OP-AMPs and derive an expression for closed loop gain of inverting configuration of OP-AMP. 6M

OR

- 10 a) Define the following Op Amp parameters: (i) CMRR, (ii) PSRR and (iii) I/P Bias Current. 6M
- b) Design an OP-AMP circuit to give an output $V_o = 3/4 V_1 + 5/6 V_2 + 6/7 V_3$, where $V_1 = 1\text{ V}$, $V_2 = 2\text{ V}$ and $V_3 = 3\text{ V}$. 8M

[B17 EC 2201]

[B17 EE 2203]
II B. Tech II Semester (R 17) Regular Examinations
CONTROL SYSTEMS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

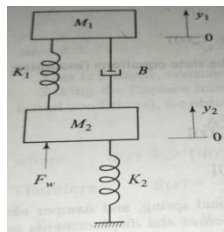
Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

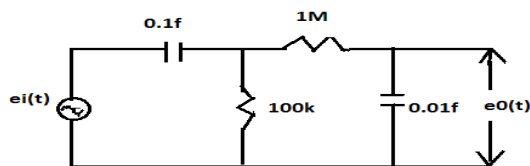
UNIT - I

- 1 a) Explain open loop and closed loop system with practical example 7M
- b) Determine the transfer function $Y_1(s)/F_w(s)$ of the system Shown in below. 7M
 Also obtain its electrical analog based on f-v analogy.



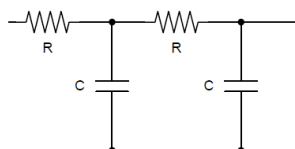
OR

- 2 a) Compare open-loop and closed loop systems. 7M
- b) Draw the block diagram representation of the network shown in fig .2 and find its transfer function $E_0(s)/E_1(s)$ using block reduction method. 7M

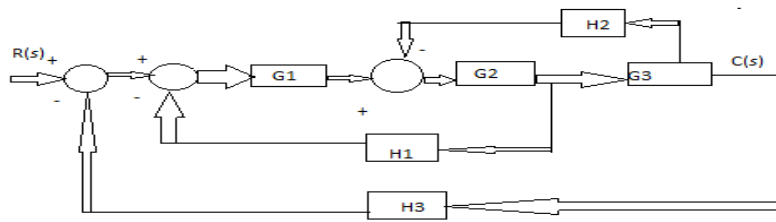


UNIT - II

- 3 a) Obtain the signal flow graph for the following electrical system and then find its transfer function using Mason's gain rule. 7M

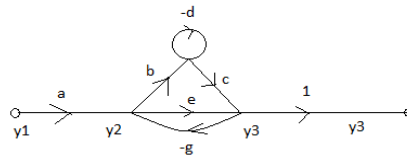


- b) Find the overall transfer function. 7M



OR

- 4 a) Find the gain of the signal flow graph shown below .where a,b,c,d,e,g and l
respective Gains of the branches. 7M



- b) Explain Mason's gain formula. 7M

UNIT - III

- 5 a) State and explain the response of a first order system when subject to step
input signal. 7M
b) A unity feedback system is characterized by an open-loop transfer function 7M
 $G(s) = K/S(S+10)$. Determine the value of gain K such that the system has a
damping ratio of 0.5. With this value of K, find the settling time and
percent overshoot of unit-step response.

OR

- 6 a) Derive the expression for unit-step response of a standard second-order 7M
system which is under-damped.
b) Determine the stability of system that has the following characteristic 7M
equation whose function is

$$G(s).H(s) = K(s+3)/s(s+2)(s+4)(s+5).$$

UNIT - IV

- 7 a) Using R-H criterion, find the range of K for the closed loop system to be stable. The open loop transfer function of the system is 7M

$$G(S)H(S) = \frac{K}{S(S+1)(S+2)}$$

- b) Determine the stability of the system that has following characteristic equation using R-H criterion. 7M

$$S^6 + 2S^5 + 8S^4 + 17S^3 + 20S^2 + 16S + 1 = 0.$$

OR

- 8 a) Obtain the Root-locus for the system with 7M

$$G(S)H(S) = \frac{K}{S(S+4)(S^2+4S+8)}.$$

What value of K makes the closed loop system marginally stable?

- b) A unity feedback system has an open loop transfer function 7M
 $G(s) = k(s+1)/s(s-1)$. sketch the root locus Plot with k as variable parameter.
 Show that the loci of complex roots are parts of a circle with (-1,0) as centre and radius $\sqrt{2}$.

UNIT - V

- 9 a) Draw the Bode plots for the system having 7M

$$G(S)H(S) = \frac{10}{S(1+0.02S)(1+0.2S)}.$$

Determine the Gain Margin and Phase Margin.

- b) Draw the polar plots for the system whose 7M

- 1) $G(S).H(S) = 1/S(1+TS)$
- 2) $G(S).H(S) = S$.

OR

- 10 a) Draw the bode plots for the transfer function 7M

$$G(s) = 50/s(1+0.25s)(1+0.1s)$$

- b) Draw Nyquist diagram and determine the stability of a control system with 7M
 open-loop transfer function $G(S)H(S) = \frac{3}{S(S+1)^2}.$

[B17 EE 2203]

[B17 EC 2202]
II B. Tech II Semester (R 17) Regular Examinations
ELECTROMAGNETIC FIELD THEORY & TRANSMISSION LINES
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT - I

- 1 a) Derive an expression for the energy stored in static electric field. 7M
- b) The point charges $2nC$, $3nC$ and $4nC$ are located at $(0,0,1)$, $(1,0,1)$ and $(1,1,1)$ respectively. Find the energy stored in the system. 7M

OR

- 2 a) Derive an expression for the electric field due to a finite line charge along z axis and hence obtain the field due to an infinite line charge along z axis. 7M
- b) State and prove uniqueness theorem. 7M

UNIT - II

- 3 a) Derive point form of Ampere's circuital law. 7M
- b) A circular loop located on $x^2 + y^2 = 9$, $z = 0$ carries a direct current of 10A in the Φ -direction. Determine magnetic field intensity **H** at $(0,0,2)$. 7M

OR

- 4 a) Derive an expression for vector magnetic potential **A** 7M
- b) A current distribution gives rise to the vector magnetic potential **A** = $xy\mathbf{a}_x + yz\mathbf{a}_y + zx\mathbf{a}_z$. Calculate **B** at $(1,2,3)$ and the flux through the surface defined by $x = 1, 0 \leq y \leq 1, 0 \leq z \leq 1$. 7M

UNIT - III

- 5 a) Derive Maxwell's equations in integral and differential form. 7M
- b) In free space, **E** = $20\sin(\omega t - 10x)\mathbf{a}_y$ v/m. Calculate displacement current density, ω and **H**. 7M

OR

- 6 a) Derive boundary conditions on **E, D, H** and **B** for time varying fields at the interface between the two media. 7M
- b) What are retarded potentials? Derive differential equations linking the retarded potentials and their sources? 7M

UNIT - IV

- 7 a) Obtain the relation between E and H in a uniform plane wave. 7M
- b) A 20 MHz uniform plane wave propagating through a lossy medium has $\beta = 0.5 \text{ rad/m}$ and its amplitude is reduced by 20% for every meter travelled. Calculate α , v and δ . 7M

OR

- 8 a) Derive expressions for the reflection coefficient for both perpendicular and parallel polarizations when a plane wave is incident obliquely on a perfect dielectric. 7M
- b) Discuss wave propagation in good dielectrics and good conductors. 7M

UNIT - V

- 9 a) Derive an expression for the input impedance of a transmission line. 7M
- b) A $75\text{-}\Omega$ lossless transmission line is terminated by load of $(100 + j100)\text{ }\Omega$. calculate reflection coefficient at load Γ_L , standing wave ratio S and input impedance Z_{in} if the line is $\lambda/4$ long. 7M

OR

- 10 a) Discuss in detail the characteristics of TE and TM waves in rectangular wave guides. 7M
- b) A rectangular waveguide measures 3×4 cm internally. A 10-GHz signal is propagated in the waveguide. Calculate cutoff wavelength, Cutoff frequency, guide wavelength, phase and group velocities and characteristic wave impedance for the dominant TE mode. 7M

[B17 EC 2202]

[B17 EC 2203]
II B. Tech II Semester (R 17) Regular Examinations
ANALOG COMMUNICATIONS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|----|------|---|----|
| 1. | (a). | Explain the AM diode detector demodulator with a neat diagram. | 7M |
| | (b). | Explain the generation of DSB-SC with the Balanced modulator method. | 7M |
| | | (OR) | |
| 2. | (a). | Explain advantages of SSB system. Describe a method of generating an SSB signal | 7M |
| | (b). | Explain Vestigial Side Band Modulation. | 7M |

UNIT-II

- | | | | |
|----|------|--|----|
| 3. | (a). | Define frequency deviation and phase deviation in Frequency Modulation and differentiate between NBFM and WBFM. | 7M |
| | (b). | In a FM system when the audio frequency is 500Hz and the AF voltage is 2.4 V, the frequency deviation is 4.8 kHz. If the AF voltage is increased to 7.2V, what is the new deviation? If the AF voltage is raised to 10V, while audio frequency is dropped to 200Hz, what is the deviation? Find the modulation index in each case. | 7M |
| | | (OR) | |
| 4. | (a). | Describe the operation of a phase locked loop FM demodulator with a neat diagram | 7M |
| | (b). | Draw the block diagram of an Indirect method of FM generation and explain. | 7M |

UNIT-III

- | | | | |
|----|------|--|-----|
| 5. | (a). | Derive the expression for signal -to-noise ratio in DSB-SC system. | 8M |
| | (b). | Explain sources of Noise | 6M |
| | | (OR) | |
| 6. | (a). | Derive the expression for signal -to-noise ratio in FM system. | 10M |
| | (b). | Explain Threshold in FM | 4M |

UNIT-IV

- | | | | |
|----|------|---|----|
| 7. | (a). | Draw the block Diagram of AM transmitter and Explain. | 8M |
| | (b). | Explain Pre-Emphasis & De-emphasis | 6M |
| | | (OR) | |
| 8. | (a). | Draw the block diagram of FM transmitter and Explain. | 7M |
| | (b). | Draw the block diagram of an SSB transmitter and Explain. | 7M |

UNIT-V

- | | | | |
|-----|------|--|----|
| 9. | (a). | Explain the principle of operation of a super-heterodyne radio receiver. | 8M |
| | (b). | What are the factors that govern the choice of Intermediate frequency? | 6M |
| | | (OR) | |
| 10. | (a). | Distinguish between AGC and Delayed AGC with help of neat circuit diagrams | 7M |
| | (b). | Define Image frequency and Image frequency Rejection Ratio. | 7M |

[B17 EC 2203]

[B17 EC 2204]
II B. Tech II Semester (R 17) Regular Examinations
COMPUTER ARCHITECTURE AND ORGANIZATION
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a). Explain the operation of 4-bit adder-subtractor with a neat diagram. 7M
(b). Design a common bus system with multiplexers to connect four 4 bit registers A,B,C,D. How this can be modified with the help of a three state buffer. 7M
(OR)
2. (a). Explain in detail the operation of one stage arithmetic logic shift unit by using function table for each input. 7M
(b). The adder- subtractor circuit has following values for input mode M and data inputs A and B. In each case, determine the values of the outputs: S_3, S_2, S_1, S_0 and C_4 . 7M

M	A	B
0	0111	0110
0	1000	1001
1	1100	1000
1	0101	1010
1	0000	0001

UNIT-II

3. (a). Explain about Timing and Control Unit design. 7M
(b). With the help of a flowchart explain the operation of a basic computer during 1. Instruction cycle 2. Interrupt cycle 7M
(OR)
4. (a). Explain in detail about basic instruction formats with examples. 7M
(b). With the help of a diagram explain the functions of the basic computer registers. 7M

UNIT-III

5. (a). Write the merits and demerits of micro programmed control unit compared to a hardwired control unit. 7M
(b). Explain the concept of microprogramming in detail with the help of an example. 7M

(OR)

6. (a). With an example, explain various addressing modes and write how to calculate the address in each case. 7M
(b). What are the characteristics of a RISC processor? What is CISC processor? 7M

UNIT-IV

7. (a). Explain Daisy Chain Interrupt Mechanism. List the advantages and disadvantages in this scheme. 7M
(b). With the help of block diagram and truth table explain the operation of an I/O interface Unit. 7M

(OR)

8. (a). Show internal configuration of DMA and explain its operation in a computer for read and write modes. 7M
(b). Explain the operation of Asynchronous Communication Interface. 7M

UNIT-V

9. (a). A system employs RAM chips of 256bytes and ROM chips of 1024bytes. It needs 2KB RAM, 4KB ROM and four interface units with four registers each and memory mapped I/O used. The two highest order bits of address bus are 00 for RAM, 01 for ROM and 10 for interface registers. Draw a memory address map for the system. 7M
(b). What is Cache memory and explain two mapping methods in implementation of Cache? 7M

(OR)

10. (a). Explain Memory hierarchy and Differentiate between Main memory and Auxiliary Memories 7M
(b). Explain the following 7M
1. Associative Memory
2. Virtual Memory

[B17 EC 2204]

[B17 BS 2201]
II B. Tech II Semester (R 17) Regular Examinations
MANAGEMENT SCIENCE
Common to ECE& EEE
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. Define Management. Explain its nature and features 14M

(OR)

2. Explain the principles of Management as outlined by Henri Fayol 14M

UNIT-II

3. Define Marketing. Explain in detail about Marketing mix. 14M

(OR)

4. What is Product life cycle? Write about Marketing Strategies based on Product life cycle 14M

UNIT-III

5. Differentiate Personnel Management from HRM 14M

(OR)

6. What are the basic functions of HR Manager? 14M

UNIT-IV

7. What is Production Planning and Control (PPC)? Explain about the functions of PPC. 14M

(OR)

8. What are the various methods of raising finance? 14M

UNIT-V

9. Explain about the importance of Mission, Goal, Objective and Strategy. 14M

(OR)

10. What do you understand by SWOT analysis? Explain how it can be carried out. 14M

[B17 BS 2201]

[B17EC3101]
III B. Tech I Semester (R17) Regular Examinations
PULSE AND DIGITAL CIRCUITS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|----|------|---|----|
| 1. | (a). | Explain the response of RC High Pass circuit for the following input waveforms
(i) Step (ii) Pulse | 7M |
| | (b). | Derive the expression for a %tilt of a square wave after passing through a high pass RC Circuit. | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 2. | (a). | Show that a high pass circuit acts as differentiator | 5M |
| | (b). | Show that a low pass circuit acts as an Integrator? | 5M |
| | (c). | Draw the circuit diagram of a compensated attenuator and derive the condition necessary for perfect attenuation. | 4M |

UNIT-II

- | | | | |
|----|------|---|----|
| 3. | (a). | State and prove clamping circuit theorem. | 7M |
| | (b). | With the help of neat Circuit diagram, explain the working of Transistor Clipper. | 7M |

(OR)

- | | | | |
|----|------|---|----|
| 4. | (a). | Design a clipping circuit to clip the given input voltage $10 \sin \omega t$ at + 5V level and below 5V. | 7M |
| | (b). | Explain the response of the clamping circuit when a square wave input is applied under steady state conditions. | 7M |

UNIT-III

- | | | | |
|----|------|---|----|
| 5. | (a). | Explain the operation of Schmitt trigger circuit with neat diagram and also derive UTP, LTP equations. | 7M |
| | (b). | Explain the operation of Collector Coupled monostable multivibrator with a circuit diagram and relevant waveforms | 7M |

(OR)

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|----|------|---|----|
| 6. | (a). | Explain the operation of Collector Coupled Astable multivibrator with a circuit diagram and relevant waveforms. | 7M |
| | (b). | Define different switching times of a transistor with suitable circuit diagram and waveforms. | 7M |

UNIT-IV

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|----|------|---|----|
| 7. | (a). | Draw a neat circuit diagram of UJT relaxation oscillator and explain its operation. Can it be used as time-base circuit of a CRO. | 7M |
| | (b). | Derive an expressions for sweep speed error in Miller and Bootstrap sweep generators. | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 8. | (a). | With the help of a neat circuit diagram and waveforms explain the working of a current Sweep generator | 7M |
| | (b). | Define and derive the expressions for the slope error, transmission error and | 7M |

displacement error for the exponential sweep circuit.

UNIT-V

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|-------------|------|---|----|
| 9. | (a). | With a neat circuit diagram explain the operation of ECL OR/NOR logic gate. | 7M |
| | (b). | Draw the circuit of a TTL NAND gate and explain the operation. | 7M |
| (OR) | | | |
| 10. | (a). | With a neat circuit diagram explain the operation of ECL OR/NOR logic gate. | 7M |
| | (b). | Draw the circuit of a CMOS inverter and explain the operation. | 7M |

[B17EC3101]

[B17EC3102]
III B. Tech I Semester (R17) Regular Examinations
LINEAR ICS AND APPLICATIONS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a). Explain the Sample and Hold Circuit with neat sketches. 7M
- (b). Draw the Instrumentation Amplifier and explain its operation in detail. 7M

(OR)

2. (a). Explain the operation of an Regenerative comparator with circuit diagram and waveforms. If $R_1 = 50 K\Omega$, $R_2 = 100 \Omega$, $\pm V_{sat} = \pm 14V$, $V_{ref} = 0.2V$ and $V_{in} = 1V(p-p)$ Find the values of V_{UT} and V_{LT} . 7M
- (b). Explain the operation of grounded load V to I converter using op-amp. 7M

UNIT-II

3. (a). Draw the 2nd order High pass filter and explain its operation in detail. 7M
- (b). Write a short note on Switched Capacitance Filters. 7M

(OR)

4. (a). Design a 1st order LPF and HPF and draw their output response. 7M
- (b). Design a Wide Band Pass Filter with $f_L = 200Hz$, $f_H = 1 KHz$ and pass band gain of Calculate the value of Q for the filter. 7M

UNIT-III

5. (a). Design RC Phase Shift Oscillator for $f = 1KHz$. 7M
- (b). Derive the frequency and the condition for sustained oscillations for Wien bridge oscillator. 7M

(OR)

6. (a). Explain about the Quadrature Oscillator. 7M
- (b). Draw and explain the functional block diagram of 566 IC 7M

UNIT-IV

7. (a). Derive Lock in range and Capture range for PLL. 7M
- (b). Explain about three terminal IC voltage regulators. 7M

(OR)

8. (a). Explain operation of Astable multi-vibrator configured around IC 555 timer and derive its frequency. 7M
- (b). Explain any two applications of PLL. 7M

UNIT-V

9. (a). List out DAC and ADC Specifications. 7M
- (b). Draw the circuit of weighted resistor DAC and derive expression for output analog voltage V_o . 7M

(OR)

10. (a). Draw and explain the working of Successive Approximation A/D converter 7M
- (b). Explain the working of Counter type A/D converter with necessary diagrams. 7M

[B17EC3102]

[B17EC3103]
III B. Tech I Semester (R17) Regular Examinations
ELECTRONIC MEASUREMENTS AND INSTRUMENTATION
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|----|------|--|----|
| 1. | (a). | Explain the different types of errors that may occur in measurements. Describe their sources and precautions to minimize them. | 7M |
| | (b). | Explain the operation of AC Voltmeter using full wave Rectifier. | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 2. | (a). | How do you extend the range of a given ammeter and voltmeter? | 7M |
| | (b). | Explain the working of a true RMS voltmeter with the help of a suitable block diagram. | 7M |

UNIT-II

- | | | | |
|----|------|---|----|
| 3. | (a). | Discuss the factors that need to be consider, while selecting a transducer? | 7M |
| | (b). | Explain Capacitive transducers in detail. | 7M |

(OR)

- | | | | |
|----|------|---|----|
| 4. | (a). | Draw the Linear variable differential Transducer and explain its working in detail. | 7M |
| | (b). | Explain about piezo electric transducer. | 7M |

UNIT-III

- | | | | |
|----|------|---|----|
| 5. | (a). | Explain the operation of Maxwell's Bridge and derive the condition for balance of a bridge. | 7M |
| | (b). | An ac bridge was made up as follows: arm ab, a capacitor of 0.8 mF in parallel with 1 kW resistance, bc a resistance of 3 kW, arm cd an unknown capacitor Cx and Rx in series, arm da a capacitance of 0.4 mF. The supply at 1 kHz is connected across bd and a detector across ac. Determine the value of unknown capacitance Cx, unknown series resistance Rx and dissipation factor. | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 6. | (a). | Explain the operation of wien's Bridge and derive the condition for balance of a bridge. | 7M |
| | (b). | The arms of a four-arm bridge <i>a</i> , <i>b</i> , <i>c</i> and <i>d</i> supplied with sinusoidal voltage have the following values.
arm <i>ab</i> : A resistance of 800 W in parallel with a capacitance of 2 mF
arm <i>bc</i> : 400 W resistance
arm <i>cd</i> : 1 kW resistance
arm <i>da</i> : A resistance <i>R</i> ₂ in series with 2 mF capacitance
Determine the value of <i>R</i> ₂ and frequency at which the bridge will balance. | 7M |

UNIT-IV

- | | | | |
|----|------|---|----|
| 7. | (a). | With a neat block diagram explain the block of a dual trace oscilloscope. | 7M |
| | (b). | Explain delay line circuit in CRO. | 7M |

(OR)

8. (a). With a neat block diagram explain Sampling oscilloscope. 7M
(b). Explain measurement of frequency by using Lissajous figure. 7M

UNIT-V

9. (a). With the block diagram explain the working of a function generator producing sine, square and triangle waveforms. 7M
(b). Explain the working of random noise generator with neat diagrams. 7M

(OR)

10. (a). Draw the block diagram of a spectrum analyzer and explain its working. 7M
(b). With a neat diagram explain the working of Square wave generator. 7M

[B17EC3103]

[B17EC3104]
III B. Tech I Semester (R17) Regular Examinations
DIGITAL COMMUNICATION
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|----|------|---|----|
| 1. | (a). | Explain the modulation and demodulation techniques for pulse time modulation systems. | 7M |
| | (b). | Explain about the operation of a PCM system. What is companding? | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 2. | (a). | Explain about delta modulation (DM) system and design an Adaptive Delta Modulation system to eliminate the drawbacks generated in DM system. | 7M |
| | (b). | The signal $v(t) = \cos 5\pi t + 0.5 \cos 10\pi t$ is instantaneously sampled. Find the maximum allowable value for sampling time T_s and write an expression for sampled signal $vs(t) = v(t)s(t)$ if the sampling signal $s(t) = 5 \sum_{k=-\infty}^{\infty} \delta(t - 0.1k)$ | 7M |

UNIT-II

- | | | | |
|----|------|--|----|
| 3. | (a). | Explain how a binary signal can be transmitted and received by using a BPSK system? | 7M |
| | (b). | With the help of an example, explain the method of generating a phase continuous MSK signal. | 7M |

(OR)

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|----|------|--|----|
| 4. | (a). | Explain the role of a QPSK transmitter and receiver in serial data transmission and reception. | 7M |
| | (b). | In a DPSK Receiver, the received bit sequence $b(t)$ is 01101100 then
i) Find reconstructed bit sequence $d(t)$
ii) Due to presence of noise $b(t)$ is recovered as 01111100 then detect $d(t)$ and identify the bits which are wrongly detected. Use EX-OR logic. | 7M |

UNIT-III

- | | | | |
|----|------|---|----|
| 5. | (a). | Derive the expression for output SNR when the binary signal is transmitted using BPSK in a PCM system. | 7M |
| | (b). | Derive the relation for output noise power spectral density as $G_{no}(f) = H(f) ^2 G_{ni}(f)$ where $G_{ni}(f)$ is input noise power spectral density of a filter with transfer function $H(f)$. | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 6. | (a). | Explain about linear filtering and calculate noise power output of RC low pass filter, differentiator and an integrator. | 7M |
| | (b). | Derive expressions for quadrature components of noise. | 7M |

UNIT-IV

- | | | | |
|----|------|---|----|
| 7. | (a). | What is the function of a base band signal receiver and derive its probability of error ? | 7M |
| | (b). | By deriving expression for P_e for BPSK and BFSK systems, compare the performance of these two data transmission systems. | 7M |

(OR)

- | | | | |
|----|------|---|----|
| 8. | (a). | Derive the expressions for Probability of error P_e and transfer function $H(f)$ for an optimum filter. | 7M |
| | (b). | A signal is either $S_1(t) = A \cos 2\pi f_0 t$ or $S_2(t) = 0$ for an interval $T = n/f_0$, with n an | 7M |

integer. Find the error probability P_e of matched filter.

UNIT-V

- | | | | |
|----|------|--|----|
| 9. | (a). | Explain in detail about Frequency Hopping spread spectrum technique. | 7M |
| | (b). | Explain the methods of generating PN sequence. | 7M |

(OR)

- | | | | |
|-----|------|--|----|
| 10. | (a). | Explain about acquisition and tracking circuit for a DS system. | 7M |
| | (b). | How a CDMA system uses spread spectrum to provide multiple access communication. | 7M |

[B17EC3104]

[B17EC3105]
III B. Tech I Semester (R17) Regular Examinations
ANTENNAS AND PROPAGATION
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a). Derive the expressions for the electromagnetic fields radiated by alternating current element. 7M

- (b). Explain the radiation mechanism of a two wire transmission line. 7M

(OR)

2. (a). Show that the radiation resistance of a quarter-wave monopole is 36.5Ω . 7M

- (b). Explain in detail the directional characteristics of dipole antennas. 7M

UNIT-II

3. (a). What are Broadside and end fire arrays? Obtain expressions for BWFN and HPBW for both types of arrays. 7M

- (b). For two element array consisting identical radiators carrying equal currents in phase, obtain positions of maxima and minima of the radiation pattern if the distance of separation $d = \lambda$. 7M

(OR)

4. (a). Explain in detail the effect of earth on vertical patterns. 7M

- (b). Two identical transmitting and receiving antennas are located at a distance of 2km. If power transmitted is 25dBm at 10GHz and received power is -65dBm, determine the gain of each antenna. 7M

UNIT-III

5. (a). Distinguish between travelling wave and standing wave antennas. 7M

- (b). Explain Current Distribution of isotropic and Omni directional antennas. 7M

(OR)

6. (a). Explain in detail the operating principles of a Helical antenna. 7M

- (b). Explain in detail about Log Periodic antenna with neat diagram. 7M

UNIT-IV

7. (a). Explain the feeding mechanisms of parabolic antennas in detail. 7M

- (b). Derive an expression for the impedance of slot antennas. 7M

(OR)

8. (a). Explain different measurement ranges in details. 7M

- (b). Explain one method of measuring gain of an antenna. 7M

UNIT-V

9. (a). Discuss ground wave propagation in detail. 7M

- (b). Derive an expression for the field strength of Space Wave. 7M

(OR)

10. (a). Explain the effect of Earth's magnetic field on ionospheric propagation. 7M

- (b). Explain in detail about MUF, duct propagation and scatter propagation. 7M

[B17EC3105]

[B17EC3106]
III B. Tech I Semester (R17) Regular Examinations
COMPUTER NETWORK ENGINEERING
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|----|------|---|----|
| 1. | (a). | Categorise the networks LAN,MAN and WAN | 7M |
| | (b). | Explain about B-ISDN ATM Reference model. | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 2. | (a). | Draw the neat diagram of ISO-OSI reference model and explain the function of each layer in detail. | 7M |
| | (b). | Explain about Topologies of networks in detail. | 7M |

UNIT-II

- | | | | |
|----|------|--|----|
| 3. | (a). | Explain about different transmission medias. | 7M |
| | (b). | Explain about Circuit, Message and Packet switching's in detail. Compare Circuit Switching and Packet switching. | 7M |

(OR)

- | | | | |
|----|------|--|----|
| 4. | (a). | What is switch hierarchy? Explain about crossbar switches. | 7M |
| | (b). | Compare TDM and FDM. | 7M |

UNIT-III

- | | | | |
|----|------|--|----|
| 5. | (a). | A bit stream 10011101 is transmitted using the standard CRC method .The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted. Suppose the third bit from the left is inverted during transmission .Show that this error is detected during transmission. | 7M |
| | (b). | Explain in detail about Go back n and Selective Repeat protocols. | 7M |

(OR)

- | | | | |
|----|------|---|----|
| 6. | (a). | Explain about pure ALOHA in detail. | 7M |
| | (b). | Explain about different types of Multiplexing's in Physical layer | 7M |

UNIT-IV

- | | | | |
|----|------|--|----|
| 7. | (a). | Explain the Leaky bucket algorithm for congestion control. | 7M |
| | (b). | Explain about IP addresses. | 7M |

(OR)

- | | | | |
|----|------|---|----|
| 8. | (a). | Explain in detail about General principles of congestion control. | 7M |
| | (b). | Explain about Token bucket algorithm. | 7M |

UNIT-V

- | | | | |
|----|------|---|----|
| 9. | (a). | What are different different elements of transport protocols and explain. | 7M |
| | (b). | Explain about UDP internet transport protocol | 7M |

(OR)

- | | | | |
|-----|------|--|----|
| 10. | (a). | Write a short note on WWW. | 7M |
| | (b). | Discuss about DNS and Electronic Mail. | 7M |

[B17EC3106]

[B17EC3201]
III B. Tech II Semester (R17) Regular Examinations
MICROPROCESSORS AND ITS APPLICATIONS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. With help of neat diagram explain the internal architecture of INTEL 8085 microprocessor 14M
(OR)

2. (a). Draw and explain the interrupt structure of 8085 microprocessor 8M
(b). Draw the timing diagram of the execution of OUT 85_H instruction 6M

UNIT-II

3. (a). Explain the addressing modes of 8085 with examples 7M
(b). write an 8085 assembly language program to multiply two data types stored in memory location 1001h and 1002h. store the result at memory location 1003h and 1004h 7M

(OR)

4. (a). What is subroutine ?explain how its implemented in 8085 using CALL and RET instructions 7M
(b). Write an 8085 assembly language program to count number of logical 1's and 0's in byte 7M

UNIT-III

5. (a). Draw and explain the functional block diagram of INTEL 8255 7M
(b). Explain the control word format and the feature of 8255 ports in various I/O modes of operation 7M

(OR)

6. (a). Draw and explain the functional block diagram of INTEL 8253 7M
(b). Write instructions to generate a 1khz square waveform counter1. Assumptions are clock frequency is 2MHZ ,the gate of counter1 is tied to +5v through a 10k Ω resistor port address for counter 0,1,2 and control word register are 80H,81H,82H AND 83H respectively. 7M

UNIT-IV

7. With neat diagram explain the internal architecture of INTEL 8086 microprocessor 14M

(OR)

8. (a). Draw the flag register of 8086 and explain the function of each flag register in details 7M
(b). Draw the interfacing of memory banks with 8086 and explain how bytes transfer and word transfer 7M

UNIT-V

9. (a). Draw the programmable register array of 8086 and explain the function of each register 7M

- (b). Write an 8086 Program to convert 8 bit BCD number stored at memory location 7M

15000H into its equivalent binary. Store the result at 15001 H memory location

(OR)

- | | | | |
|-----|------|---|----|
| 10. | (a). | Explain the data addressing modes of 8086 with examples | 7M |
| | (b). | Write an 8086 program to add four digit BCD numbers present in memory locations 15000 H and 15002 H. Store the result at memory location 15004 H. | 7M |

[B17EC3201]

[B17EC3202]
III B. Tech II Semester (R17) Regular Examinations
MICROWAVE ENGINEERING
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks:70

Answer ONE QUESTION from each unit.

All questions carry equal marks.

UNIT-I

1. (a). Explain the operation of a Magic Tee and its applications in detail. 7M
- (b). State the principle of operation of Directional coupler. Explain the operation of two hole directional coupler in detail. 7M

(OR)

2. (a). 1. An air filled rectangular cavity resonator has $a=d=2\text{cm}$, and $b=1\text{cm}$ and is operated in the TE_{101} mode calculate 7M
 - I. Resonant frequency.
 - II. If the cavity is filled with a dielectric of a relative permittivity 2.5, what is the resonant frequency?
- (b). State Faraday rotation principle. Explain the operation of isolator. 7M

UNIT-II

3. (a). What is a scattering matrix? Write the properties of a scattering matrix 7M
- (b). Derive the S-parameters for Magic Tee. 7M

(OR)

4. (a). Show that the 'S' matrix of a perfectly matched 2-port network is $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$. 7M
- (b). Explain the operations of directional coupler with the help s-parameters 7M

UNIT-III

5. (a). Explain the limitations of conventional tubes at Microwave frequencies in detail 7M
- (b). Explain the working of Reflex klystron with neat diagram 7M

(OR)

6. (a). Explain the working of Helix travelling wave tube with neat diagram 7M
- (b). Explain the working of Magnetron with neat diagram 7M

UNIT-IV

7. (a). Explain in detail the principle of operation of GUNN diode and detail different modes of operation of gunn diode. 7M
(b). Explain the operation IMPATT diode with suitable diagrams. 7M

(OR)

8. (a). Explain the operation TRAPATT diode with suitable diagrams 7M
(b). Explain the operation TUNNEL diode with suitable diagrams 7M

UNIT-V

9. (a). Explain the procedure with a neat diagram to measure the coupling factor and directivity of a given directional coupler. 7M
(b). Explain the procedure for measurement of low and high VSWR with block diagram. 7M

(OR)

10. Draw the Block Diagram of Microwave bench setup and explain the each block 14M

[B17EC3202]

[B17EC3203]
III B. Tech II Semester (R17) Regular Examinations
VLSI DESIGN
(Common to ECE & EEE(Open Elective))
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|-------------|------|---|----|
| 1. | (a). | Derive the expression for the ratio between $Z_{p.u}$ and $Z_{p.d}$ if an NMOS inverter is to be driven from another NMOS inverter. | 7M |
| | (b). | Explain the p-well process for a CMOS inverter with neat diagrams. | 7M |
| (OR) | | | |
| 2. | (a). | With neat sketches explain NMOS transistor fabrication approach? | 8M |
| | (b). | Determine Transconductance and Figure of merit of MOS transistor? | 6M |

UNIT-II

- | | | | |
|-------------|------|---|----|
| 3. | (a). | Design a stick and Layout diagrams for NMOS shift register cell. | 7M |
| | (b). | Discuss in detail the stick diagram design style of CMOS with neat diagrams of a single bit shift register cell. | 7M |
| (OR) | | | |
| 4. | (a). | Discuss in detail the stick diagram design style of NMOS for 4:1 inverter. | 7M |
| | (b). | What are the design rules? State Lambda based design rules for the widths and separation of conducting paths and for contact cuts and vias. | 7M |

UNIT-III

- | | | | |
|-------------|------|---|----|
| 5. | (a). | Define and explain sheet resistance and calculate ON resistance for 4:1 NMOS inverter? | 7M |
| | (b). | Explain the limitations on scaling of MOS circuit for substrate doping and Miniaturization? Discuss the limits of scaling due to Subthreshold currents. | 7M |
| (OR) | | | |
| 6. | (a). | Explain different scaling factors for device parameters? | 7M |
| | (b). | Explain area capacitance of layers and the calculations of wiring capacitances in a MOS transistor? | 7M |

UNIT-IV

- | | | | |
|-------------|------|--|----|
| 7. | (a). | Explain the basic architecture of FPGA. | 7M |
| | (b). | Discuss briefly about the FPGA design flow process. | 7M |
| (OR) | | | |
| 8. | (a). | Write short notes on practical design for test guidelines. | 7M |
| | (b). | With the help of an example explain built in self-test. | 7M |

UNIT-V

- | | | | |
|-------------|------|--|-----|
| 9. | (a). | Briefly explain Deep submicron digital IC design flow | 7M |
| | (b). | Explain the power consumption in low power CMOS circuits | 7M |
| (OR) | | | |
| 10. | | Write about Grid and Clock optimization of switching activity in CMOS circuits | 14M |

[B17EC3203]

[B17EC3204]
III B. Tech II Semester (R17) Regular Examinations
DIGITAL SIGNAL PROCESSING
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. (a). Find the Z-transform of the signal $x(n) = 2^n u(n) - 3^n u(-n - 1)$ and its region of convergence 6M
- (b). Realise the series & parallel canonical realizations of the following digital transfer function 8M

(OR)

2. (a). Compute the response of the system $y(n) = 0.7y(n - 1) - 0.12y(n - 2) + x(n - 1) + x(n - 2)$ To the input $x(n) = u(n)$, Discuss the stability of the above DT system? 7M
- (b). Find the inverse Z transform of $X(z) = \frac{z^2}{z^2 - 2rz \cos \theta + r^2}$ 7M

UNIT-II

3. (a). Compute the DFT of the following sequence using Radix-2 DIT FFT flow graph. Show the all intermediate stage results: $x(n) = \{0, 1, 2, 0, -2, -1, 0, 0\}$ 7M
- (b). Find the DFT of the sequence $x(n) = \{3, 2, 5, 4\}$, Using this result, find the DFT of $\{25, 20, 15, 10\}$. State the property of DFT used? 7M

(OR)

4. (a). Obtain the circular convolution of the two sequences given below using DFT method $x_1(n) = \{1, -2, 3, 1\}$, $x_2(n) = \{2, 3, 0, -4\}$ 7M
- (b). Compare inplace computation and natural input-natural output computation methods. Discuss the computational complexities involved in direct DFT and FFT. 7M

UNIT-III

5. (a). Compare Chebyshev and Butterworth analog filters ? 6M
- (b). Design digital Butterworth lowpass IIR filter using BLT method. The filter specifications are given by i) -3dB cutoff frequency at 0.5π rad, ii) atleast 15dB attenuation at 0.75π rad 8M

(OR)

6. (a). Compare Impulse invariance and Bilinear transformation methods of IIR digital filter design. 9M
- (b). Convert the following analog filter with transfer function using impulse invariance method 5M

UNIT-IV

7. (a). Design a linear-phase low pass FIR digital filter to meet the following specifications: (i) Pass band = 0 to 10 kHz (ii) Sampling frequency = 100 kHz (iii) Filter order = 10. Compute the impulse response of the desired FIR digital filter using Hamming window 6M

- (b). What is Gibb's phenomenon? Discuss the selection criteria of windows with respect to FIR filter design. 8M

(OR)

8. (a). Show that FIR filters provide constant group delay and phase delay? 7M
(b). Design a linear-phase band pass FIR digital filter to meet the following specifications: 7M
(i) Pass band = 100Hz to 200Hz
(ii) Sampling frequency = 1000Hz.
(iii) No. of samples = 11. Compute the impulse response of the desired FIR digital filter using Rectangular and Hamming windows.

UNIT-V

9. (a). Explain how Sub band coding of speech signals reduces the bit rate 7M
(b). Illustrate the operation of up-sampler, down-sampler, Interpolator and Decimator in time and frequency domains with neat sketches 7M

(OR)

10. (a). What is a digital filter bank? Describe its operation & applications. 7M
(b). Write short notes on the finite precision arithmetic effects in the realization of digital filters. 7M

[B17EC3204]

[B17EC3205]
III B. Tech II Semester (R17) Regular Examinations
RADAR ENGINEERING
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

- | | | |
|-----------------|--|-----|
| 1. | a. Draw the block diagram of a basic Radar system and explain its operation. | 7M |
| | b. Discuss the system losses in Radar briefly. | 7M |
| OR | | |
| 2. | a. Define minimum Detectable signal give the factors that affecting the minimum detectable signal. | 10M |
| | b. Write about different radar applications. | 4M |
| UNIT-II | | |
| 3. | a. Explain the Principle of operation of an MTI radar with neat schematic diagram. | 7M |
| | b. Give the various limitations to the MTI radar performance. | 7M |
| OR | | |
| 4. | a. Write about Delay line Cancellers. | 7M |
| | b. Explain about Moving Target Detector. | 7M |
| UNIT-III | | |
| 5. | a. Discuss the Airborn Doppler Radar. | 7M |
| | b. Draw the block diagram of a Monopulse Tracking Radar system and explain its operation. | 7M |
| OR | | |
| 6. | a. Explain about Sequential lobing and Conical Scan. | 7M |
| | b. Write short note on ECM and ECCM. | 7M |
| UNIT-IV | | |
| 7. | a. Write about Noise figure and Noise Temperature. | 7M |
| | b. What is a Duplexer? Explain different types of Duplexers. | 7M |
| OR | | |
| 8. | a. What is a Mixer? Explain different types of Mixers? | 7M |
| | b. Write about Radar Displays. | 7M |
| UNIT-V | | |
| 9. | a. Illustrate the working of ILS briefly? | 7M |
| | b. Describe the salient features of TACAN. | 7M |
| OR | | |
| 10. | a. Write short note on OMEGA and Radio Altimeter. | 7M |
| | b. Write about principles of Direction Finders and sense finders. | 7M |

[B17EC3205]

[B17EC3206]
III B. Tech II Semester (R17) Regular Examinations
MICROCONTROLLERS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. With the help of neat diagram, explain the architecture of 8051. 14M

(OR)

2. Explain the memory organisation of 8051. 14M

UNIT-II

3. What are the different addressing modes available in 8051 and list out bit manipulation instructions with examples? 14M

(OR)

4. Explain the instruction syntax of 8051 and write a program to transfer block of data from internal memory locations 20h to 30h to external memory locations 3020h to 3030h. 14M

UNIT-III

5. What is the difference between timer and counter? Explain the functionality of each bit in TMOD and TCON registers. 14M

(OR)

6. What is an interrupt? Write an assembly program to generate a waveform with ON time of 7 msec and OFF time of 21msec on p0.5. Assume XTAL of 11.0592MHz. Use timer 0. 14M

UNIT-IV

7. Interface 4x4 keypad to 8051 and explain how scanning and identifying the key pressed is done. Write an ALP to interface 4x4 keypad to 8051. 14M

(OR)

8. Draw and explain interfacing diagram of ADC with 8051 microcontroller. 14M

UNIT-V

9. Explain the Architecture of 8255A. Write an ALP to transfer letter 'Y' serially at 9600 baud rate continuously. 14M

(OR)

10. Explain about serial communication along with the steps required for programming 8051 to transfer and receive data serially. 14M

[B17EC3206]

[B17CS3214]
III B. Tech II Semester (R17) Regular Examinations
OOPS THROUGH JAVA
(Common to ECE & EEE)
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|----|----|--|----|
| 1. | a. | What are the oops concepts explain with example program. | 7M |
| | b. | Explain different types of primitive data types. | 7M |

OR

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|----|----|--|----|
| 2. | a. | What is type conversion and casting? Explain in detail. | 7M |
| | b. | Explain in detail about identifiers, literals and operators. | 7M |

UNIT-II

- | | | | |
|----|----|---|----|
| 3. | a. | Explain methods, classes, messages. | 7M |
| | b. | Define overloading and exceptions? Explain in detail. | 7M |

OR

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|----|----|---|----|
| 4. | a. | Explain class declaration and creating objects with an example. | 7M |
| | b. | Write about the importance of static keyword. | 7M |

UNIT-III

- | | | | |
|----|----|--|----|
| 5. | a. | What is super keyword and final keyword explain. | 7M |
| | b. | Write the importance of CLASSPATH in java. | 7M |

OR

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|----|----|--|----|
| 6. | a. | What is user defined exceptions explain with an example. | 7M |
| | b. | Briefly define about try, catch, throw and assertions. | 7M |

UNIT-IV

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|----|----|--|----|
| 7. | a. | What are the difference between multitasking and multithreading. | 7M |
| | b. | Define thread? Explain thread life cycle in detail. | 7M |

OR

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|----|----|--|----|
| 8. | a. | Write about the reading data from files and writing data into the files. | 7M |
| | b. | What is random access files? Explain. | 7M |

UNIT-V

- | | | | |
|----|----|--|----|
| 9. | a. | Define Applet. What are the concepts of Applet? | 7M |
| | b. | How to create an Applet explain with an example. | 7M |

OR

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|-----|----|---|----|
| 10. | a. | Define swing. Explain in detail.. | 7M |
| | b. | What is component and container in swing. | 7M |

[B17CS3214]

[B17CS3215]
III B. Tech II Semester (R17) Regular Examinations
DATA MINING
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|----|----|--|----|
| 1. | a. | What is data mining? In how many ways the data can be mined explain. | 7M |
| | b. | What is data quality? Discuss. | |

OR

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|----|----|--|----|
| 2. | a. | Discuss about the major issues in data mining. | 7M |
| | b. | What is data visualization, data similarity and dissimilarity. | 7M |

UNIT-II

- | | | | |
|----|----|---|----|
| 3. | a. | What are the steps involved in data pre-processing. | 7M |
| | b. | Explain in detail about data cleaning. | 7M |

OR

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|----|----|--|----|
| 4. | a. | What are the different types of data transforming techniques are there? Explain. | 7M |
| | b. | Explain about data integration.. | 7M |

UNIT-III

- | | | | |
|----|----|--|----|
| 5. | a. | What is decision tree? Explain how it works for classification problem.. | 7M |
| | b. | What are the general approaches for solving classification problem? | 7M |

OR

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|----|----|---|----|
| 6. | a. | Explain the different methods for expressing an attribute test conditions | 7M |
| | b. | Draw and explain the algorithm for decision tree induction. | 7M |

UNIT-IV

- | | | | |
|----|----|--|----|
| 7. | a. | Discuss the Naïve Bayesian Classification with an example. | 7M |
| | b. | Define the Bayes Theorem and explain for classification. | 7M |

OR

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|----|----|---|----|
| 8. | a. | Discuss the FP growth algorithm with an example. | 7M |
| | b. | Write the Apriori algorithm for frequent item set generation. | 7M |

UNIT-V

- | | | | |
|----|----|---|----|
| 9. | a. | What is cluster analysis? Explain different types of clusters with neat sketch. | 7M |
| | b. | Write and explain the Agglomerative hierarchical clustering algorithm. | 7M |

OR

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|-----|----|--|----|
| 10. | a. | Write the basic K-means algorithm and explain. | 7M |
| | b. | Explain the DB Scan clustering. | 7M |

[B17CS3215]

[B17ME3210]
III B. Tech II Semester (R17) Regular Examinations
INDUSTRIAL ROBOTICS
(Common to ECE & EEE)
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

1. a. Define the terms 'Robot' and 'Robotics'. Discuss the role of robots in engineering. 7M
- b. How do you specify a robot? Is robotics automation? Discuss the different classification systems of robots. 7M

OR

2. a. How can we classify the robotics by coordinate systems, Explain. 7M
- b. Write an over view of robotics with present and future applications. 7M

UNIT-II

3. a. Apply a suitable technique for the two degree of freedom for both Rotational and Linear system and derive the equation of motion. 7M
- b. Explain about the function line diagram representation of robot arms. 7M

OR

4. a. Classify the different types of arms of robot, and explain in detail each. 7M
- b. Write in detail about locomotion devices. 7M

UNIT-III

5. a. Write about the different problems of homogeneous transformations of rotation. 7M
- b. Explain about D-H notation joint coordinates and world coordinates. 7M

OR

6. a. Define and derive the Newton-Euler formulations in robotics. 7M
- b. Briefly discuss about Lagrange-Euler notations in robotics. 7M

UNIT-IV

7. a. Explain how the trajectory planning can be performed in robotics. 7M
- b. Write about skew motion and joint integrated motion. 7M

OR

8. a. What are the general considerations are there in path description? Explain. 7M
- b. Discuss about the different types of software packages in robot programming. 7M

UNIT-V

9. a. Discuss the applications and working principle of the following sensors. 7M
i) Range sensors ii) Acoustic sensors iii) Tactile sensors
- b. What are the different actuators used in the robots? Describe them briefly. 7M

OR

10. a. Write about the applications of robots in manufacturing. 7M
- b. Explain about the importance of motors in robotics. 7M

[B17ME3210]

[B17EE3209]
III B. Tech II Semester (R17) Regular Examinations
POWER ELECTRONICS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | |
|----|---|----|
| 1. | (a). Derive Two transistor analogy of SCR | 7M |
| | (b). Explain Static characteristics of SCR. | 7M |

(OR)

- | | | |
|----|--|----|
| 2. | (a). Explain Dynamic characteristics of SCR | 7M |
| | (b). Explain the series and parallel operation of SCR. | 7M |

UNIT-II

- | | | |
|----|--|----|
| 3. | (a). Explain R and RC Triggering Circuits. | 7M |
| | (b). Explain Class-B Commutation | 7M |

(OR)

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|----|--------------------------------------|----|
| 4. | (a). Explain UJT Triggering circuit. | 7M |
| | (b). Explain Class D Commutation | 7M |

UNIT-III

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|----|---|----|
| 5. | (a). Explain the operation of single-phase half-controlled converter with RL-Load.
Derive the expression for output voltage. | 7M |
| | (b). Explain the operation of Dual Converter. | 7M |

(OR)

- | | | |
|----|---|----|
| 6. | (a). Explain the operation of single-phase full-controlled converter with RL-Load.
Derive the expression for output voltage. | 7M |
| | (b). For a 3-Ø fully controlled SCR bridge converter operating from 3-Ø ,400 volts,50 hz supply.Calculate the average D.C output voltage for a firing angle of 45°. | 7M |

UNIT-IV

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|----|--|----|
| 7. | (a). Explain step up Cycloconverter with a circuit diagram and neat output voltage waveforms. | 7M |
| | (b). Draw the circuit diagram and output voltage waveform of 1-Ø A.C voltage controller with RL load.Derive the expression for output voltage. | 7M |

(OR)

- | | | |
|----|--|----|
| 8. | (a). Explain the operation of step up chopper and Derive the expression for output voltage. | 7M |
| | (b). Explain the operation of step-down chopper and Derive the expression for output voltage | 7M |

UNIT-V

- | | | |
|----|--|----|
| 9. | (a). Explain the three phase VSI with 180° operation with neat output voltage waveforms. | 7M |
|----|--|----|

- (b). Give the Comparison of Voltage Source Inverters and Current source Inverters. 7M
- (OR)**
10. (a). Explain the three phase VSI with 120° operation with neat output voltage waveforms. 7M
- (b). Explain Pulse Width Modulation 7M

[B17EE3209]

[B17EC3207]
III B. Tech II Semester (R17) Regular Examinations
BIO MEDICAL ENGINEERING
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|-------------|------|---|----|
| 1. | (a). | Write in detail about spectral estimation in biomedical signals | 7M |
| | (b). | Enumerate the origin of bio potentials with example. | 7M |
| (OR) | | | |
| 2. | (a). | What is meant by Joint probability? | 7M |
| | (b). | Write in detail about objectives of biomedical signal analysis | 7M |

UNIT-II

- | | | | |
|-------------|------|--|----|
| 3. | (a). | Discuss the electric activity of the heart. What is the significance of the Einthoven's triangle? | 7M |
| | (b). | Explain the use of computers in analysis of biomedical signals. | 7M |
| (OR) | | | |
| 4. | (a). | Explain how time frequency analysis is helpful in biomedical signal processing. | 7M |
| | (b). | Design an optimal filter to remove noise from a signal, given that the signal and noise processes are independent, stationary, random processes. | 7M |

UNIT-III

- | | | | |
|-------------|--|--|-----|
| 5. | (a). | write in detail about turning point algorithm | 7M |
| | (b). | Write in detail about frequency domain analysis of ECG | 7M |
| (OR) | | | |
| 6. | After applying the AZTEC algorithm to a signal, the saved data array is
{2, 50, -4, 30, -6, 50, -6, 30, -4, 50, 2, 50}. | | 14M |
| | (i) Draw the waveform that AZTEC would reconstruct from these data. | | |
| | (ii) What is the amount of data reduction? | | |
| | (iii) What is the peak-to-peak amplitude of a signal reconstructed from these data? | | |

UNIT-IV

- | | | | |
|-------------|------|---|----|
| 7. | (a). | Comment on ICA. How is it applicable for cocktail party like problem as applied to EEG signals | 7M |
| | (b). | Explain the principles of Adaptive noise canceller with an example. | 7M |
| (OR) | | | |
| 8. | (a). | Given a biomedical signal, identify discrete signal epochs and correlate them with events in the related physiological processes. | 7M |
| | (b). | Write a note on Joint Time-frequency analysis of biomedical signals. | 7M |

UNIT-V

- | | | | |
|----|------|---|----|
| 9. | (a). | Write down the classification of EEG rhythms based on the frequency bands | 7M |
| | (b). | Mention the types of artifacts interfering with the EEG acquisition and the method by which they are processed. | 7M |

(OR)

10. (a). Propose a method to detect the presence of the α rhythm in an EEG channel. How is it extended to detect the presence of the same rhythm simultaneously in two channels? 7M
- (b). With a suitable algorithm, substantiate the adaptive segmentation of EEG signals. 7M

[B17EC3207]

[B17CS3216]
III B. Tech II Semester (R17) Regular Examinations
ARTIFICIAL NEURAL NETWORKS
ELECTRONICS AND COMMUNICATION ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 70 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

UNIT-I

- | | | | |
|---|----|---|----|
| 1 | a. | Explain in detail about biological neural network. | 7M |
| | b. | Explain basic building blocks of Artificial neural network. | 7M |

OR

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|---|----|--|----|
| 2 | a. | Explain different types of neuron activation functions. Describe the role of bias in activation. | 6M |
| | b. | Distinguish between supervised and unsupervised learning. | 8M |

UNIT-II

- | | | | |
|---|----|--|----|
| 3 | a. | Describe the Analysis of Pattern Mapping Networks | 6M |
| | b. | What is Bidirectional Associative Memory? Give its structure and also its algorithm. | 8M |

OR

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|---|--|--|-----|
| 4 | | Write about in detail pattern recognition in neural network. | 14M |
|---|--|--|-----|

UNIT-III

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|---|----|--|----|
| 5 | a. | Describe the analysis of pattern storage networks. | 7M |
| | b. | What are the stochastic networks? How are they different from other types of network | 7M |

OR

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|---|--|---|-----|
| 6 | | Briefly explain the following:
i) Learning vector quantization ii) Adaptive pattern classification. | 14M |
|---|--|---|-----|

UNIT-IV

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|---|----|--|----|
| 7 | a. | Write algorithm for Hebb rule for pattern Association | 7M |
| | b. | Write algorithm for Delta rule for pattern Association | 7M |

OR

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|---|----|-----------------------------------|----|
| 8 | a. | Explain about ART Algorithm. | 6M |
| | b. | Explain about Pattern Visibility. | 8M |

UNIT-V

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|---|--|---|-----|
| 9 | | Write different applications of Image processing based on ANN | 14M |
|---|--|---|-----|

OR

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|----|--|----------------------------------|-----|
| 10 | | Write about optimization in ANN. | 14M |
|----|--|----------------------------------|-----|

[B17CS3216]